



US012398004B2

(12) **United States Patent**
Shepherd

(10) **Patent No.:** **US 12,398,004 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **SUPPORT APPARATUS FOR PROPPANT STORAGE CONTAINERS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 338 days.

(21) Appl. No.: **18/136,854**

(22) Filed: **Apr. 19, 2023**

(65) **Prior Publication Data**

US 2023/0249926 A1 Aug. 10, 2023

Related U.S. Application Data

(62) Division of application No. 17/086,050, filed on Oct. 30, 2020, now Pat. No. 11,661,291.
(Continued)

(51) **Int. Cl.**
B65G 65/00 (2006.01)
B65D 88/54 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **B65G 65/005** (2013.01); **B65D 88/54**
(2013.01); **B65G 11/026** (2013.01); **B65G 65/40** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC B65G 65/005; B65G 11/026; B65G 2203/0258; B65G 65/40; B65G 2203/042; B65D 88/54
(Continued)

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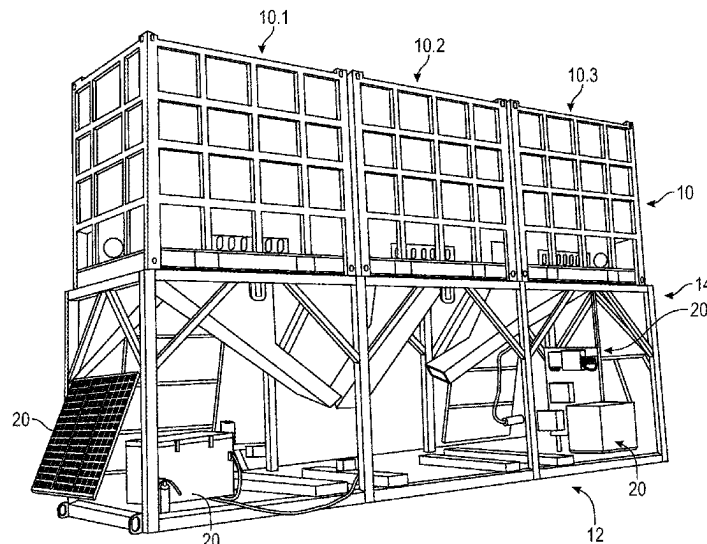
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(57) **ABSTRACT**

A support apparatus includes a frame assembly with an elevated load surface for supporting a modular proppant container in a position above a ground level. A chute assembly is supported by the frame assembly beneath the elevated load surface. A gate actuator has a coupling configured to engage with a gate assembly of the modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly. The support apparatus may include a base frame section having a recessed region beneath the elevated load surface providing a feed station, an on-board subsystem attached to the frame assembly for operating the support apparatus in a stand-alone mode or an in-situ weigh station configured to measure the weight of the modular proppant container supported on an elevated surface in the rectangular container bay.

5 Claims, 21 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 62/928,784, filed on Oct. 31, 2019.
- (51) **Int. Cl.**
B65G 11/02 (2006.01)
B65G 65/40 (2006.01)
- (52) **U.S. Cl.**
 CPC *B65G 2203/0258* (2013.01); *B65G 2203/042* (2013.01)
- (58) **Field of Classification Search**
 USPC 177/145
 See application file for complete search history.

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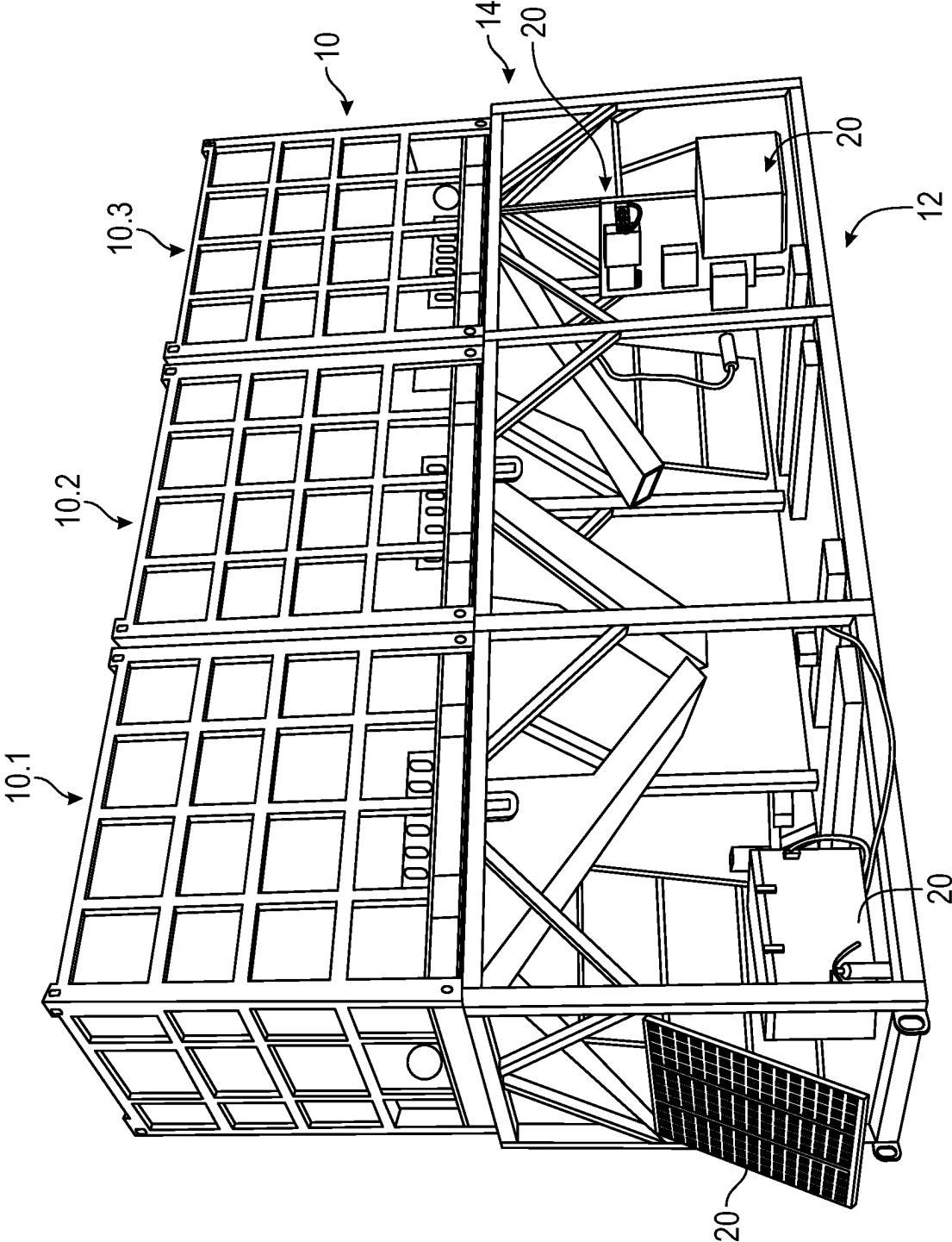


FIG. 1

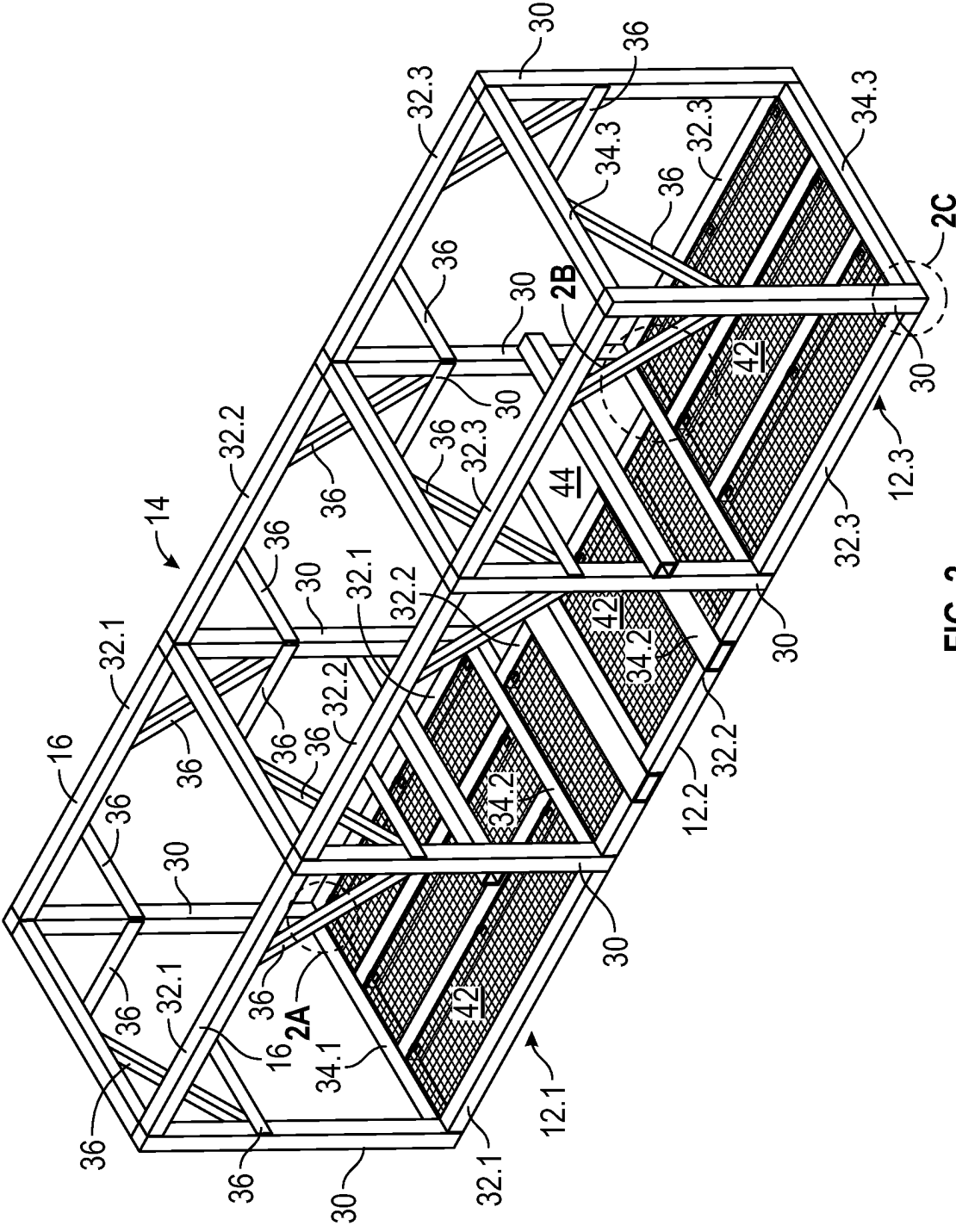


FIG. 2

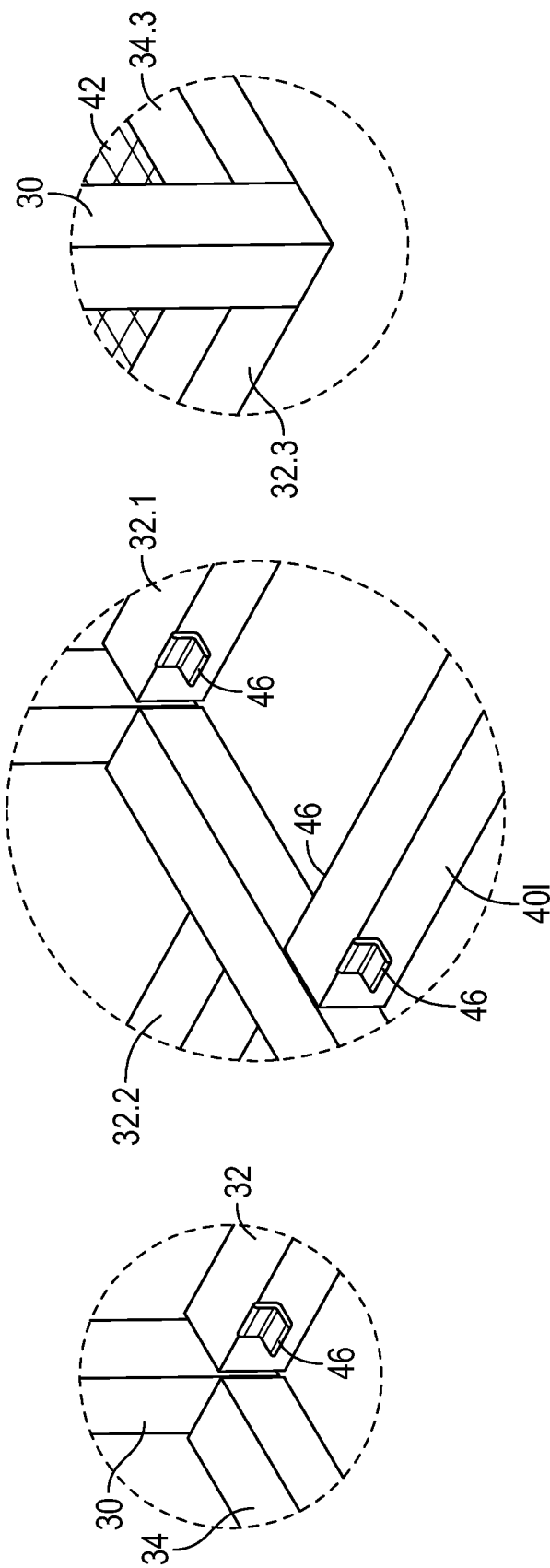


FIG. 2A

FIG. 2B

FIG. 2C

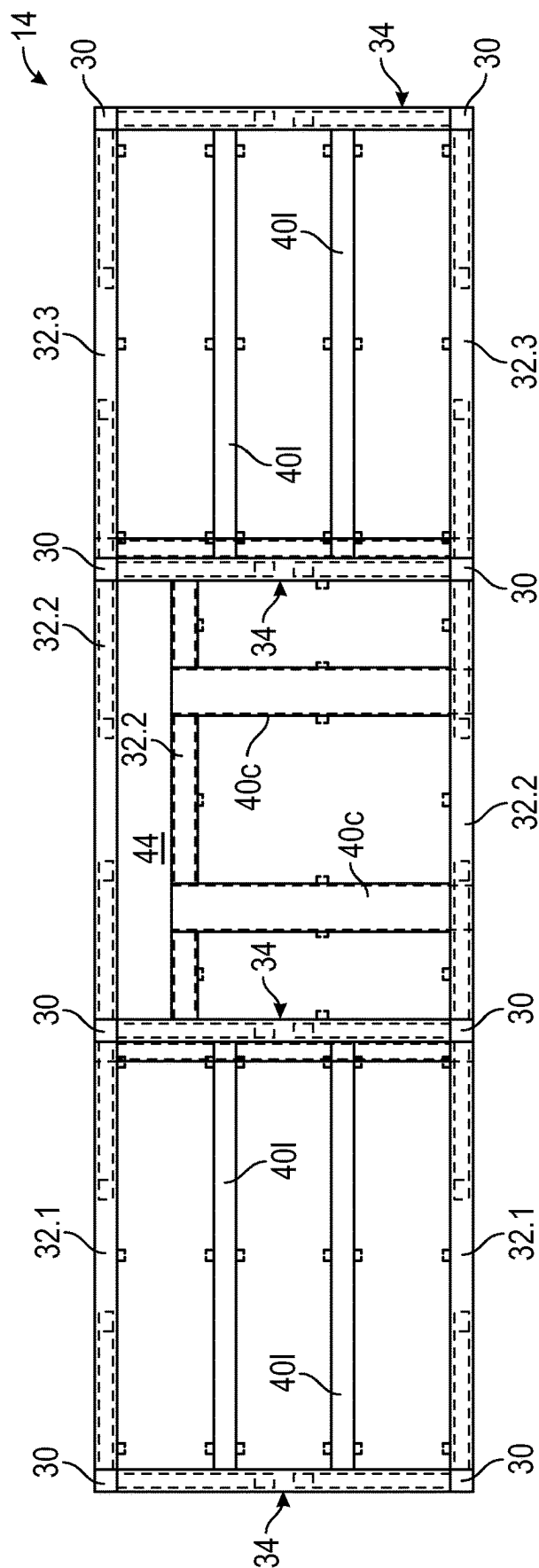


FIG. 3

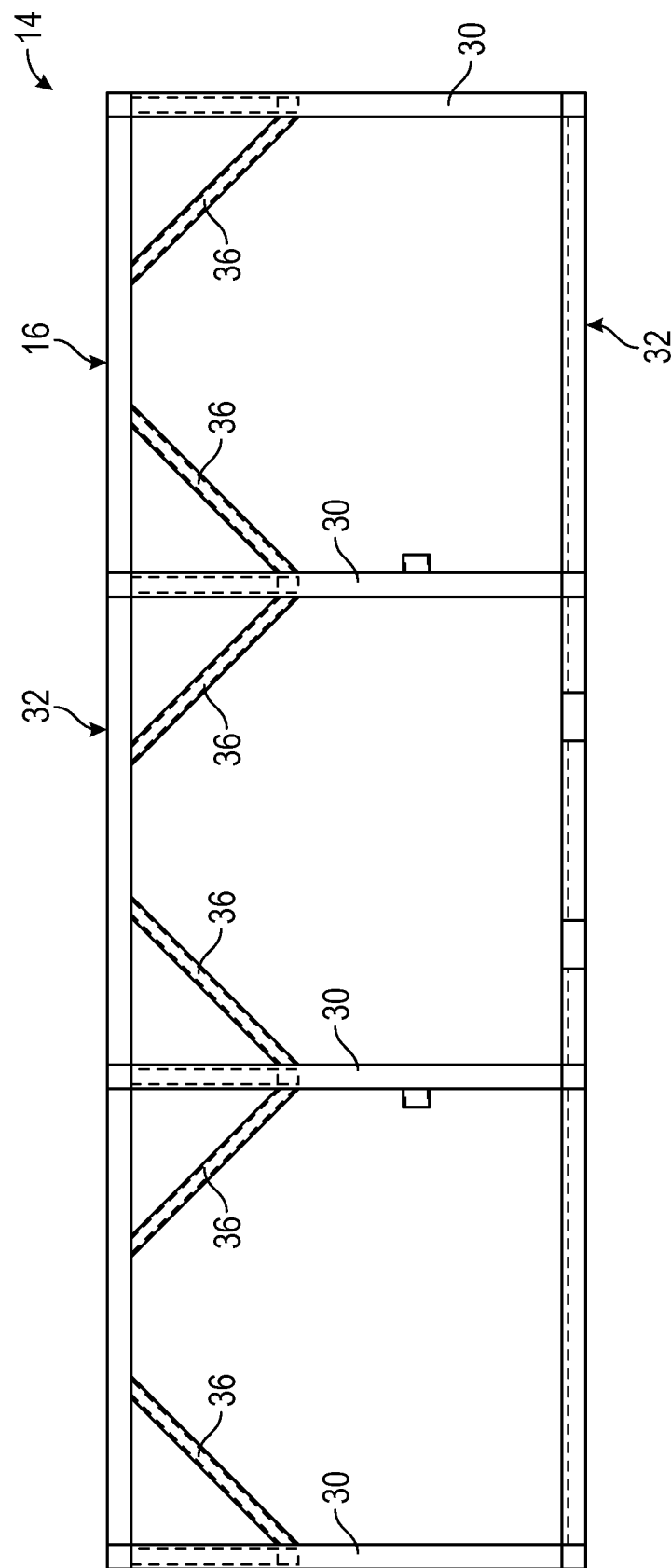


FIG. 4

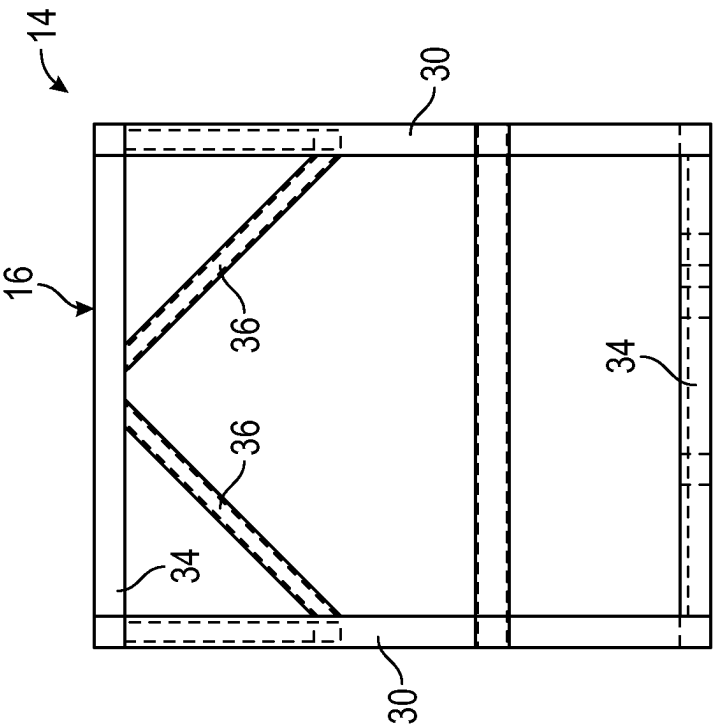


FIG. 5

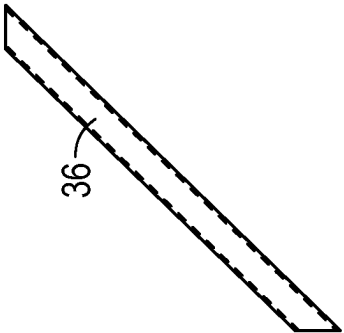


FIG. 6

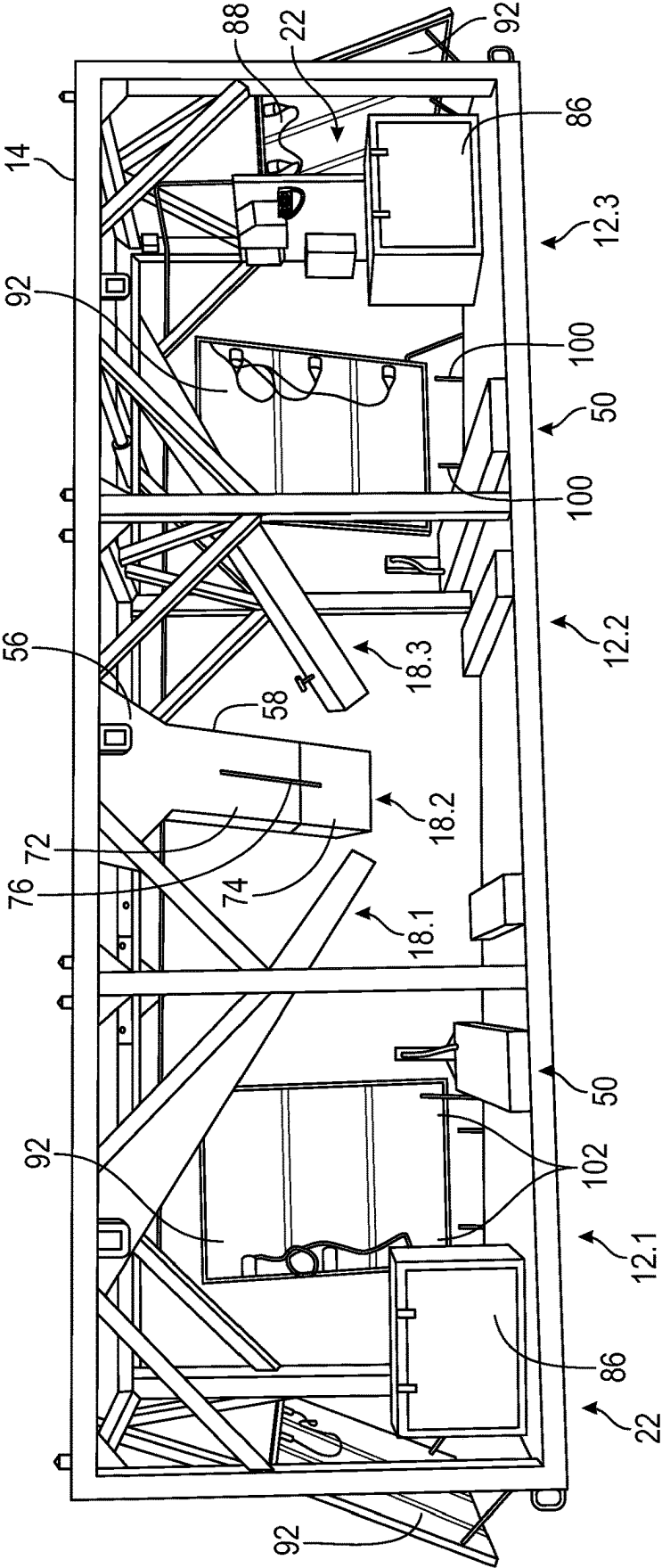
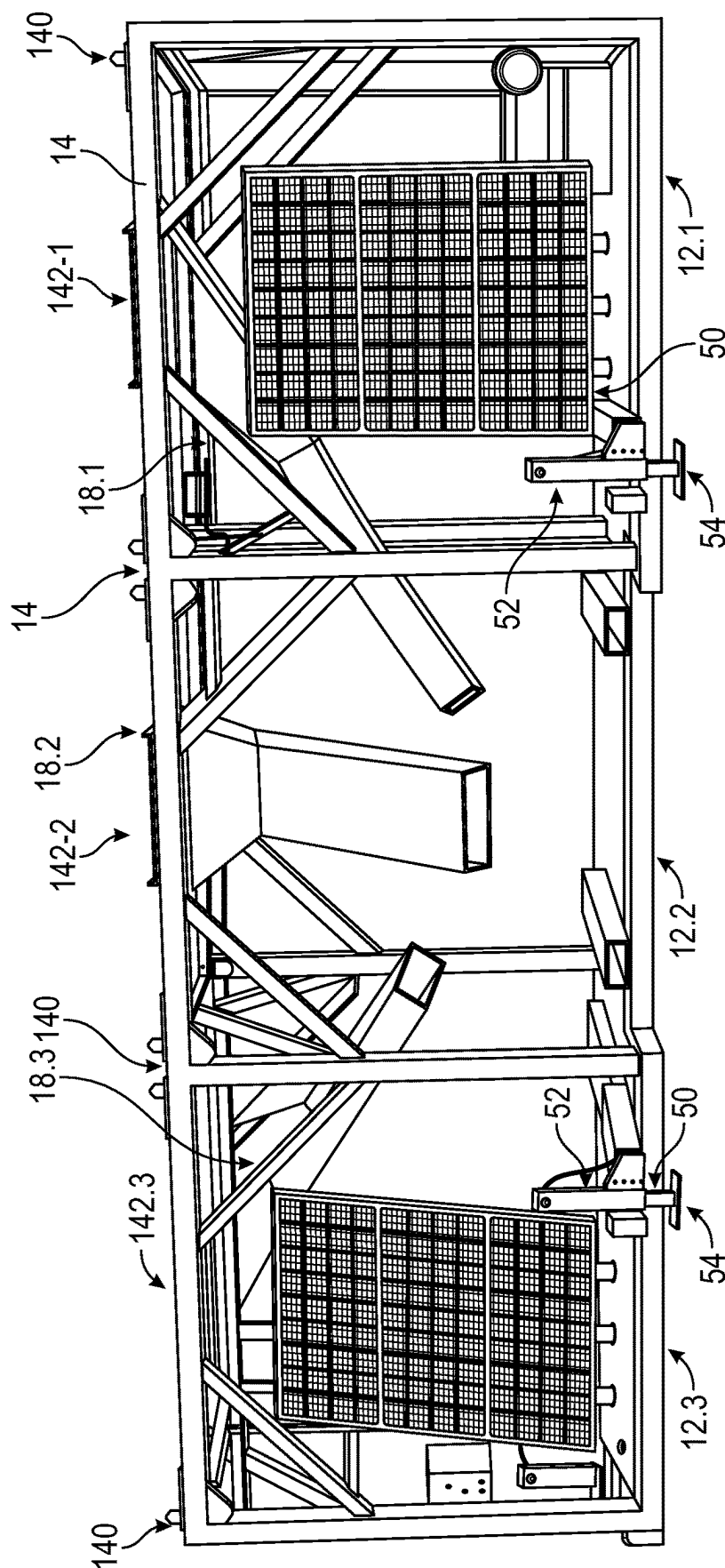


FIG. 7



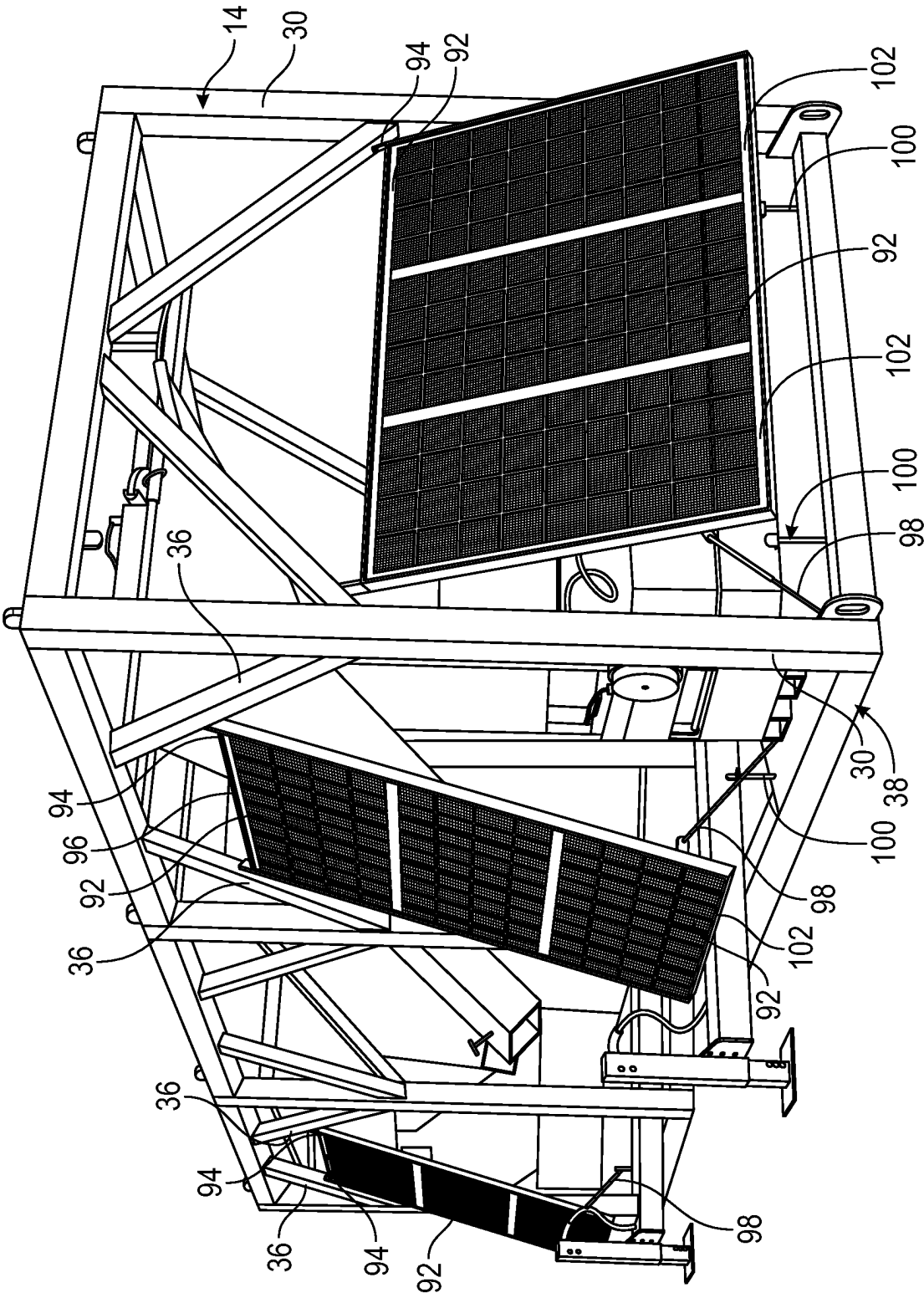


FIG. 9

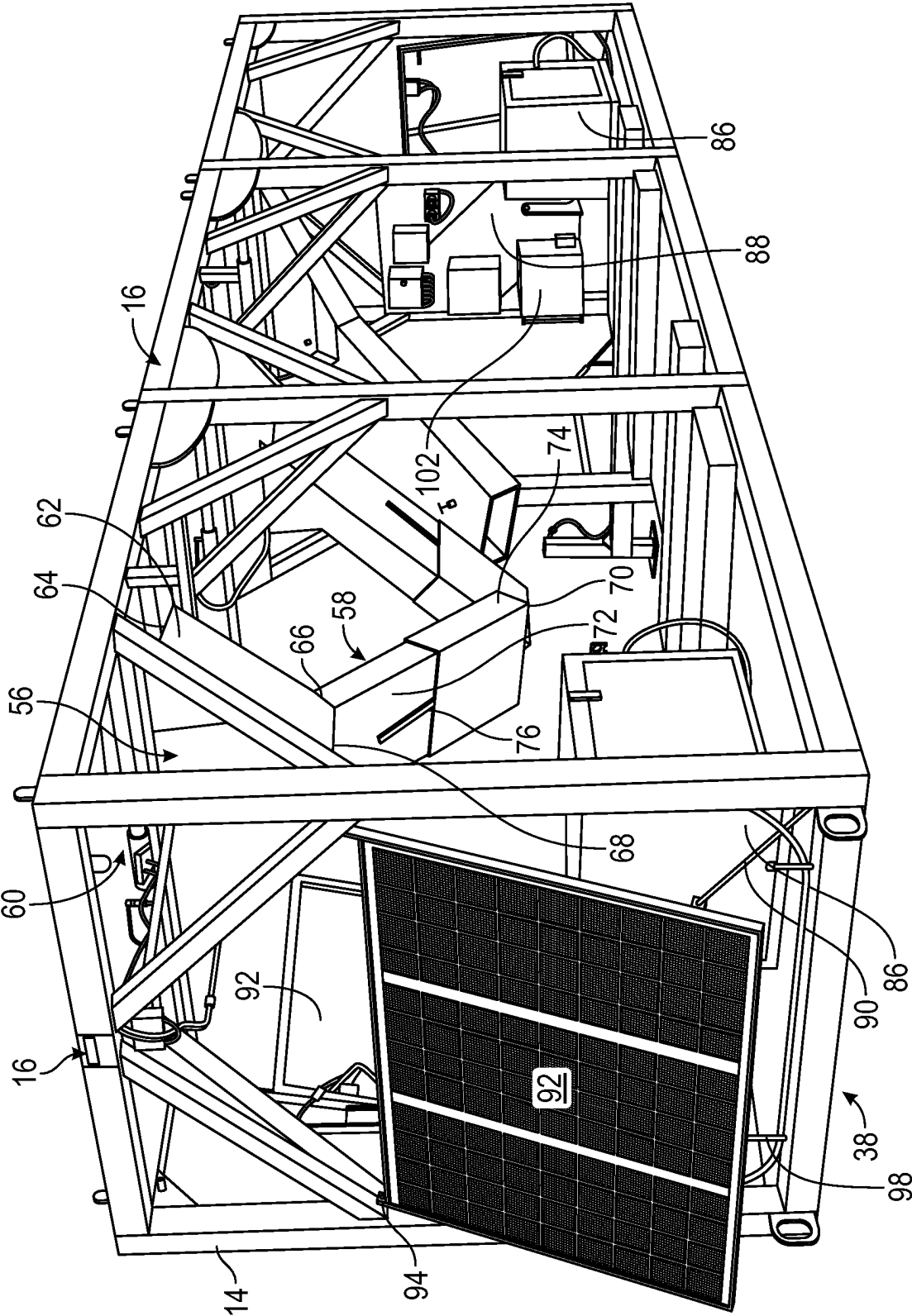


FIG. 10

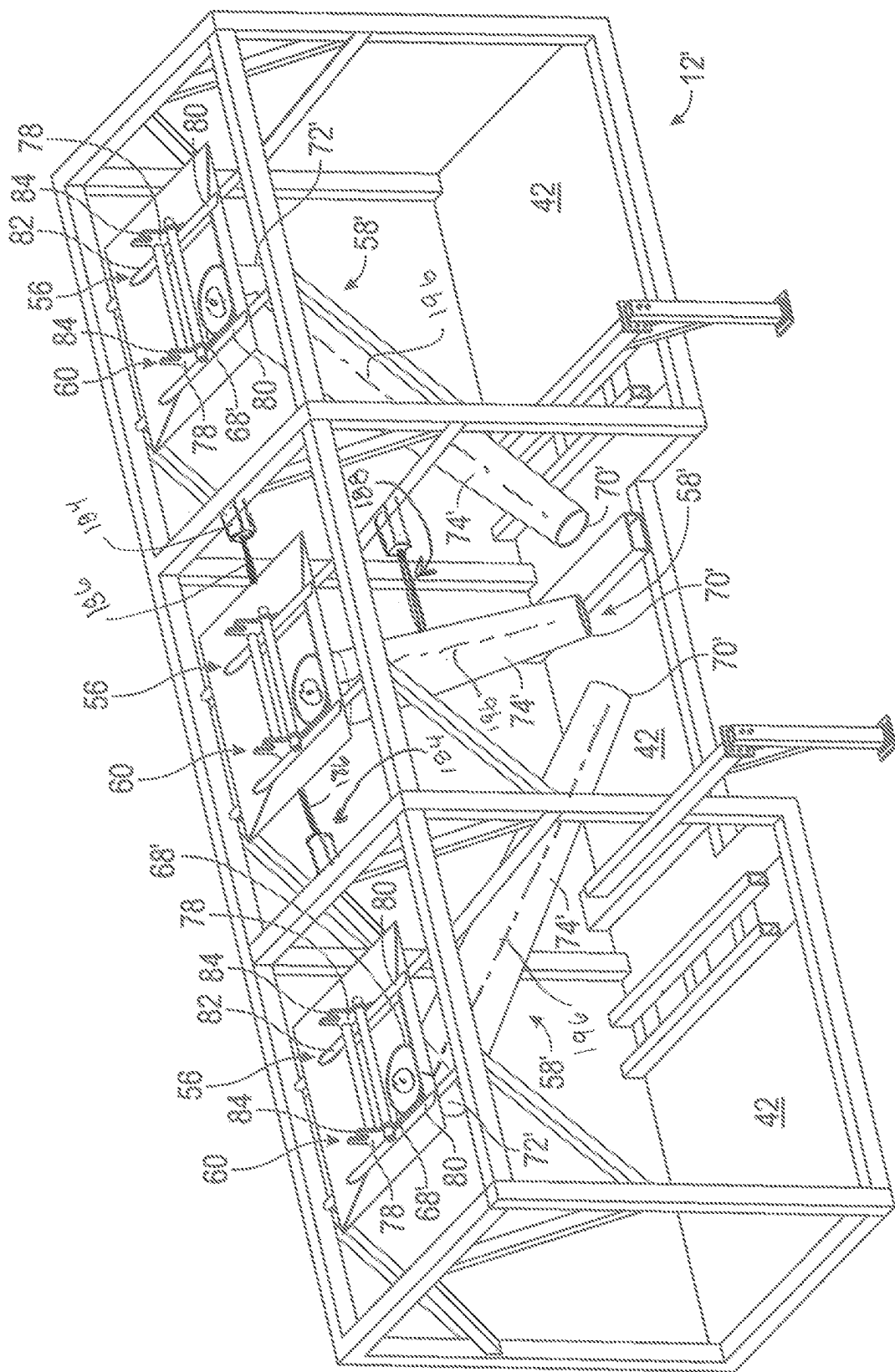


FIG. 11

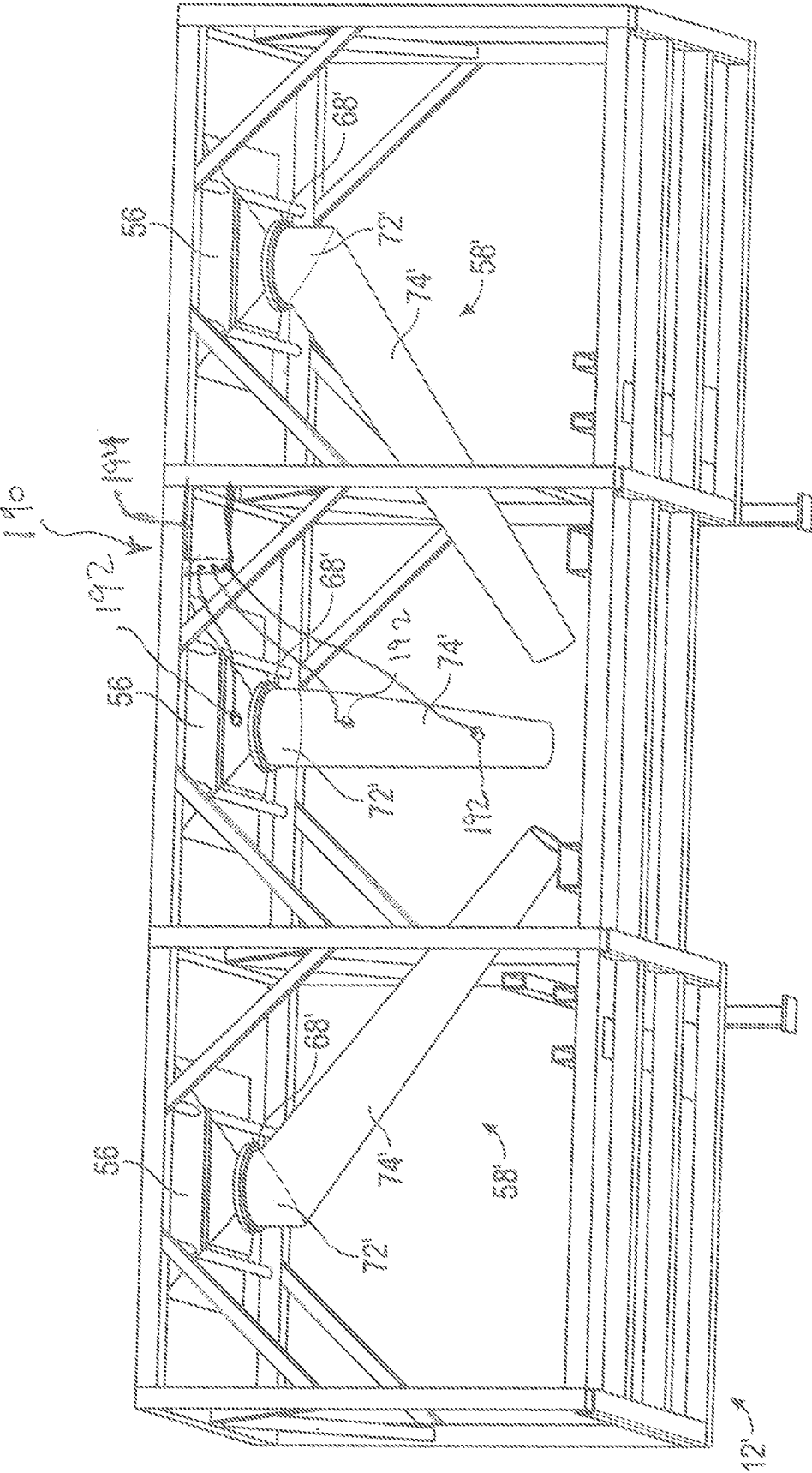


FIG. 12

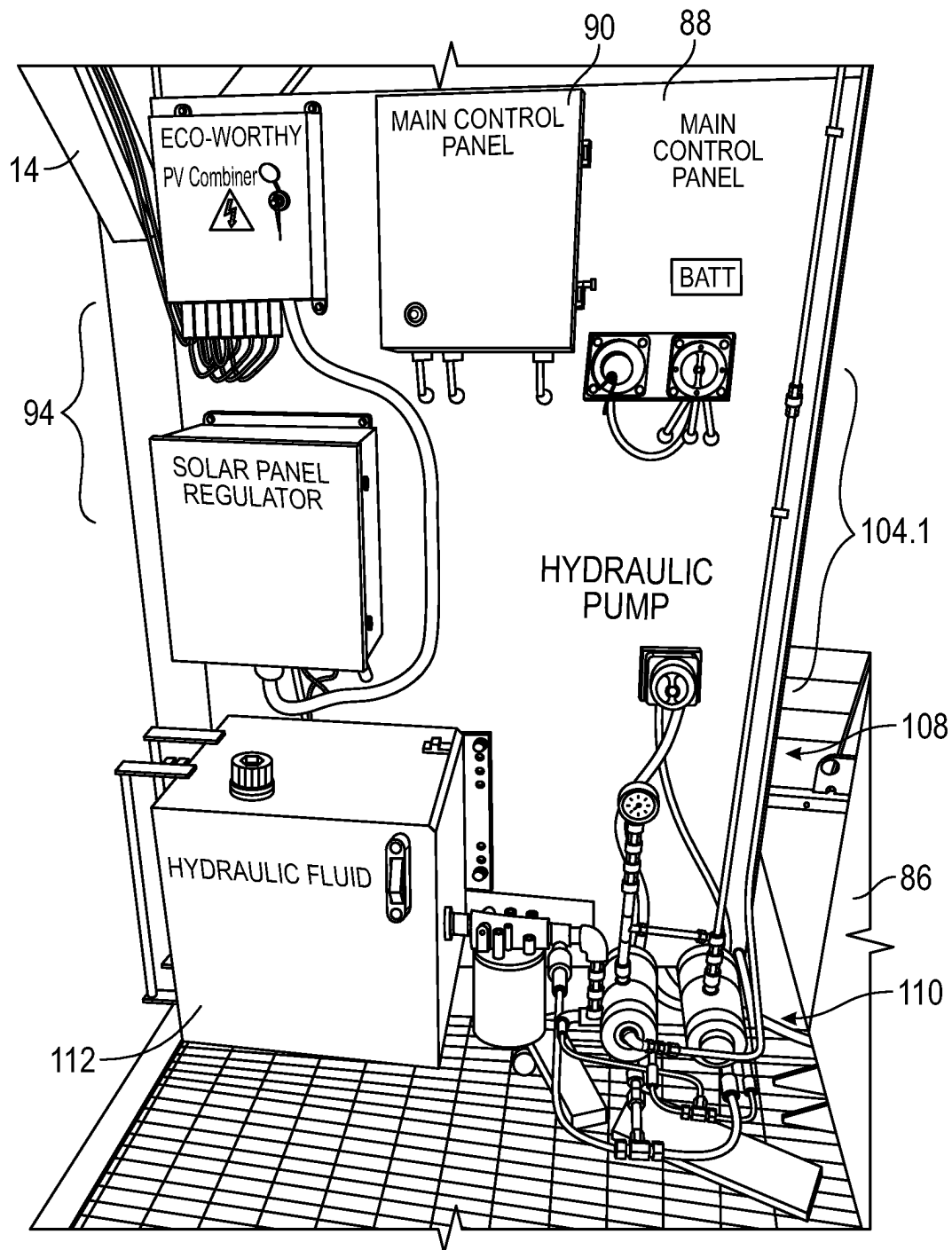


FIG. 13

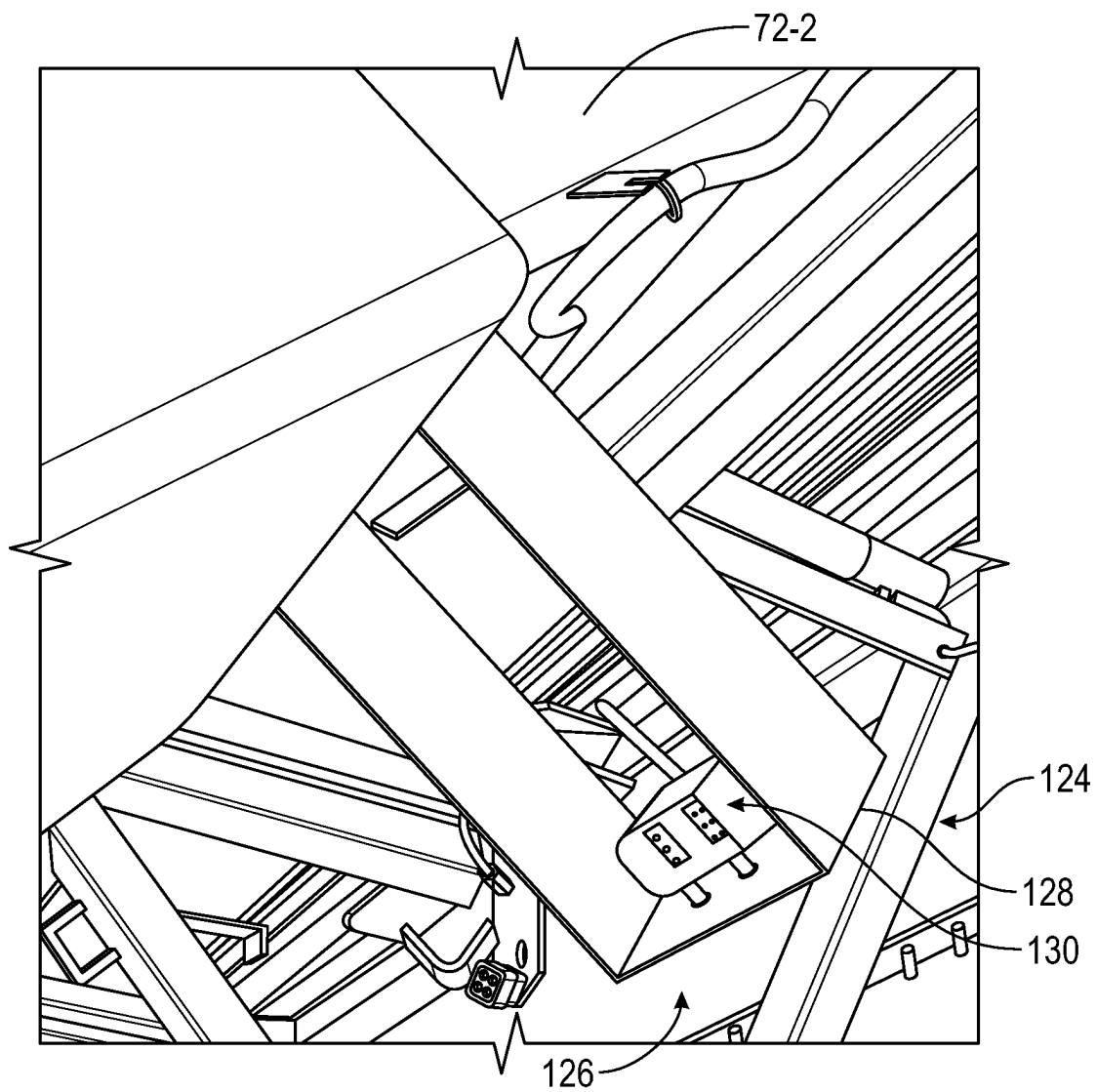


FIG. 14

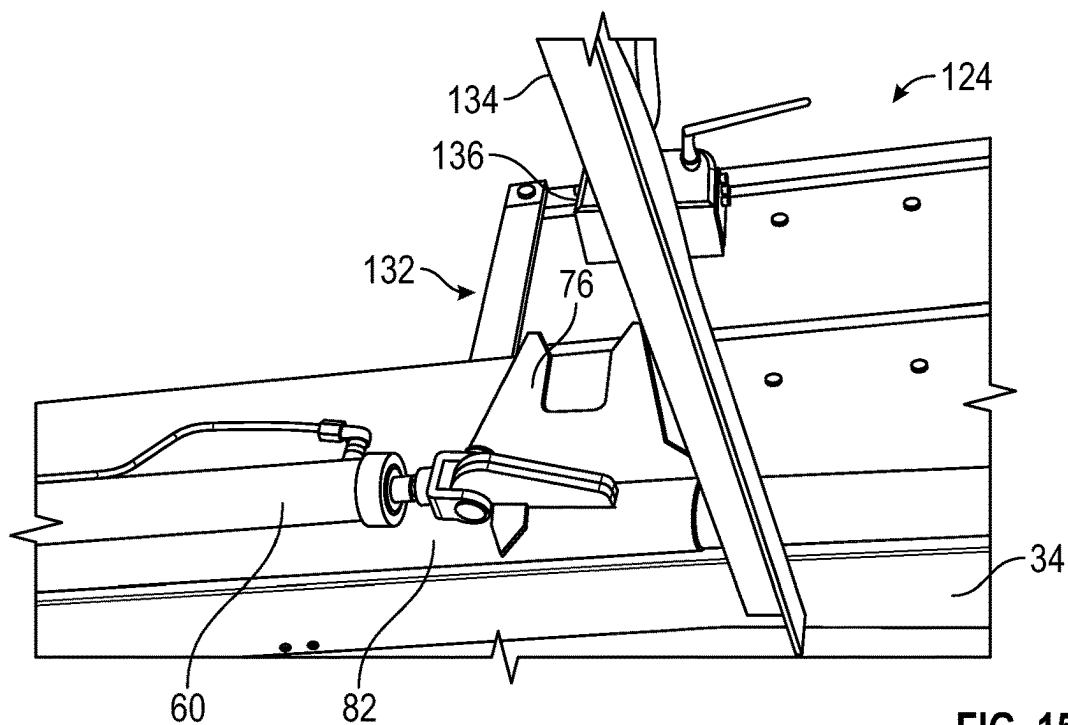


FIG. 15

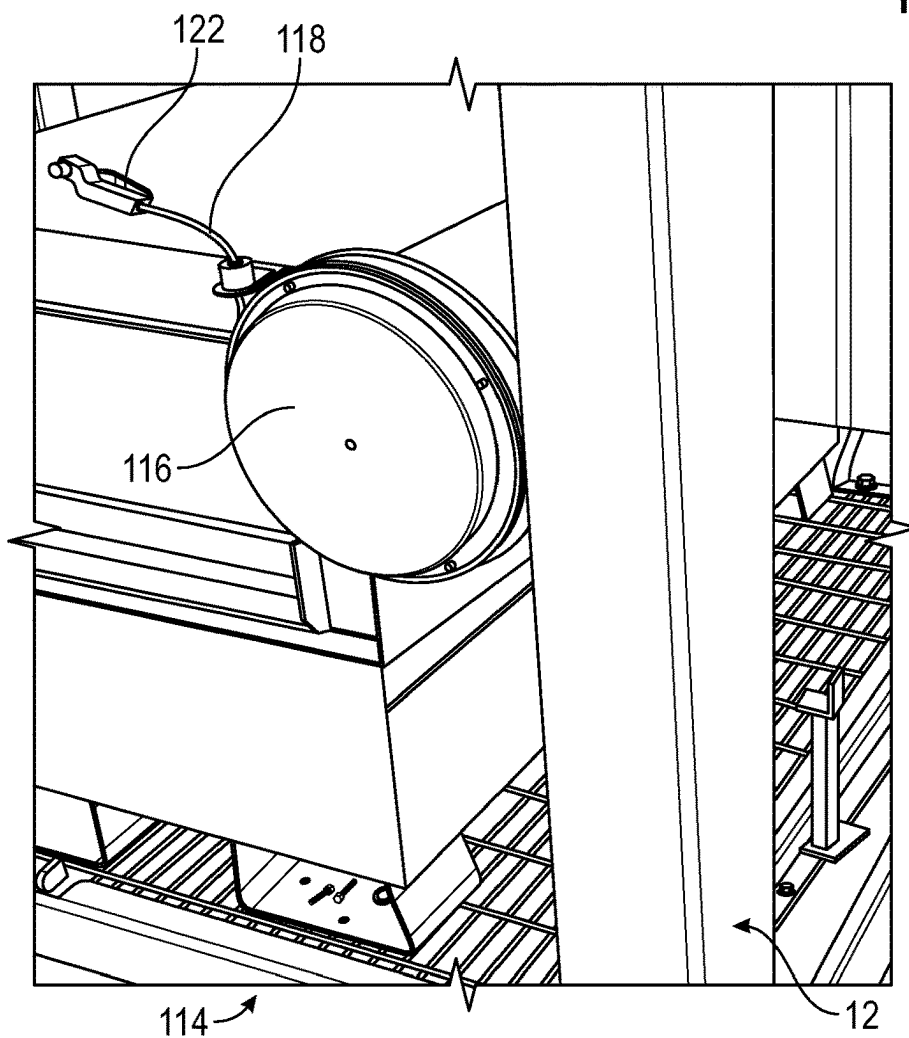


FIG. 16

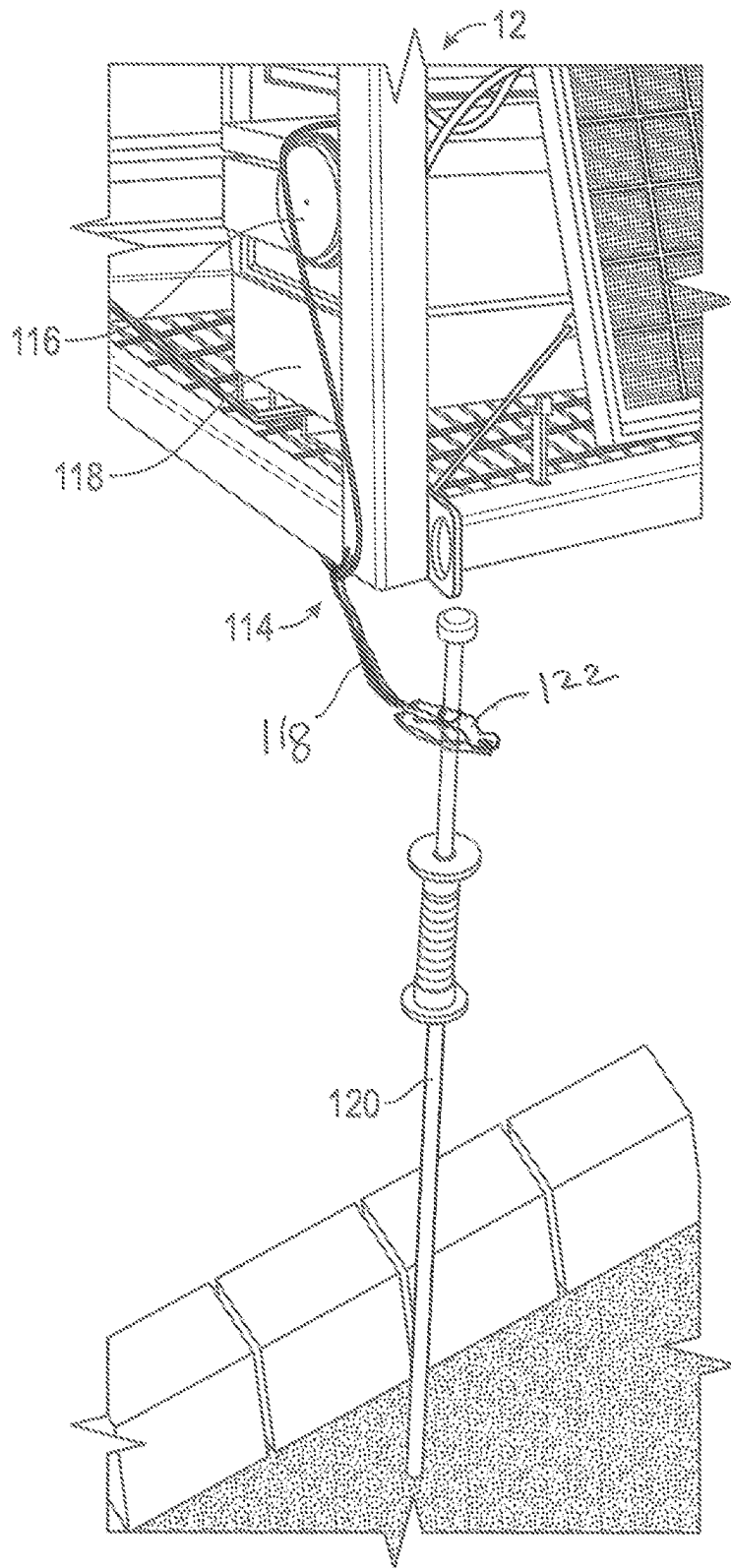


FIG. 17

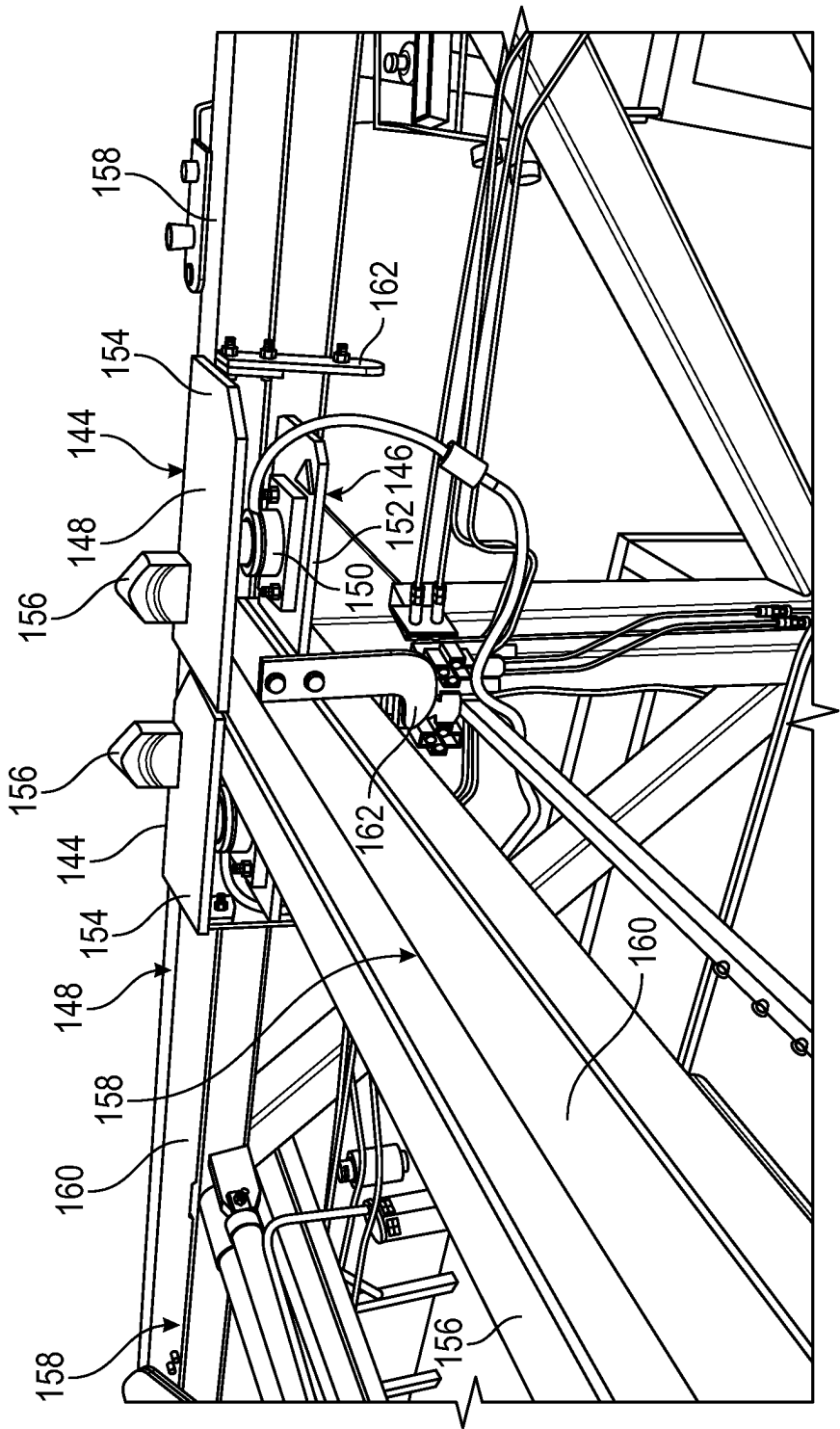


FIG. 18

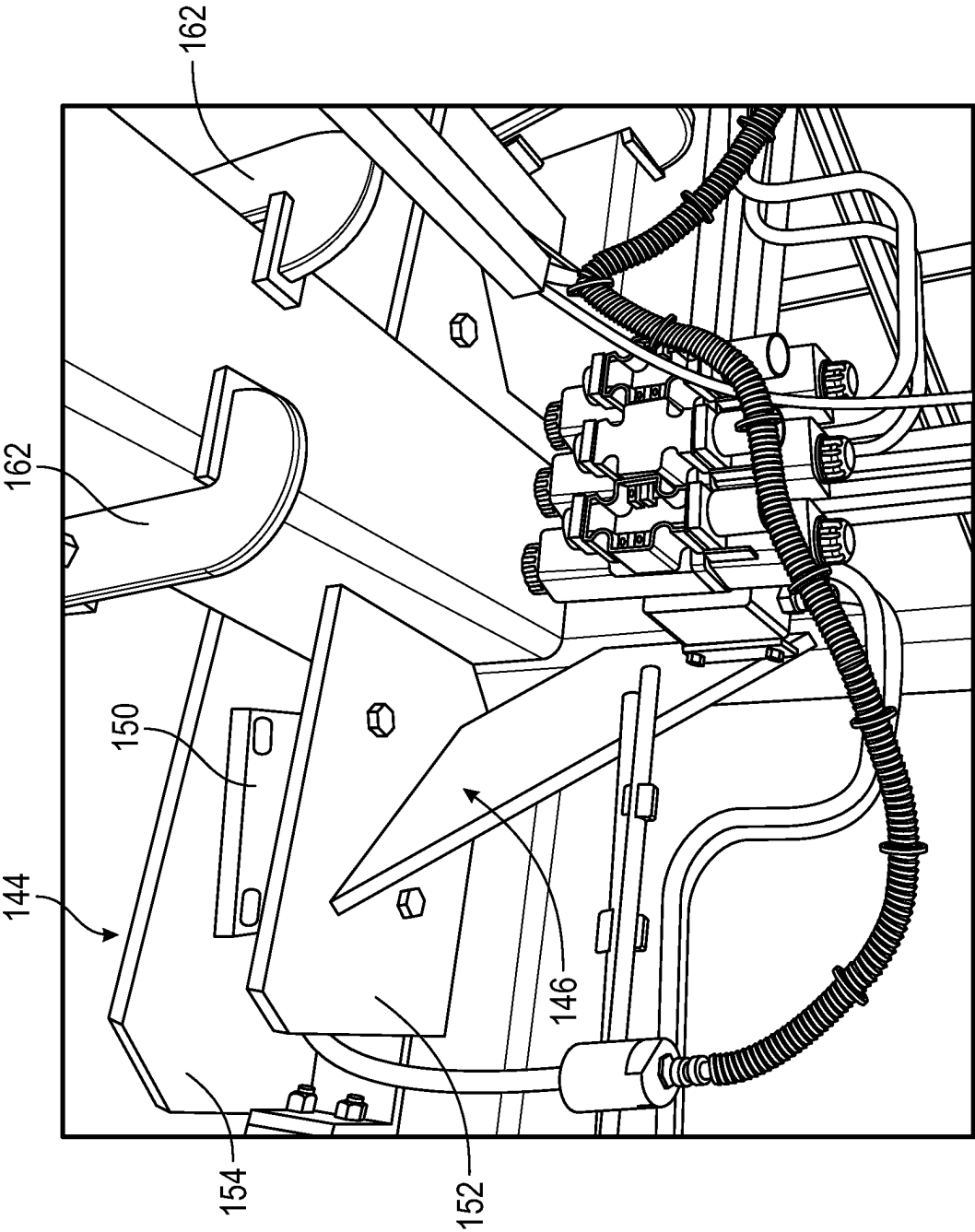


FIG. 19

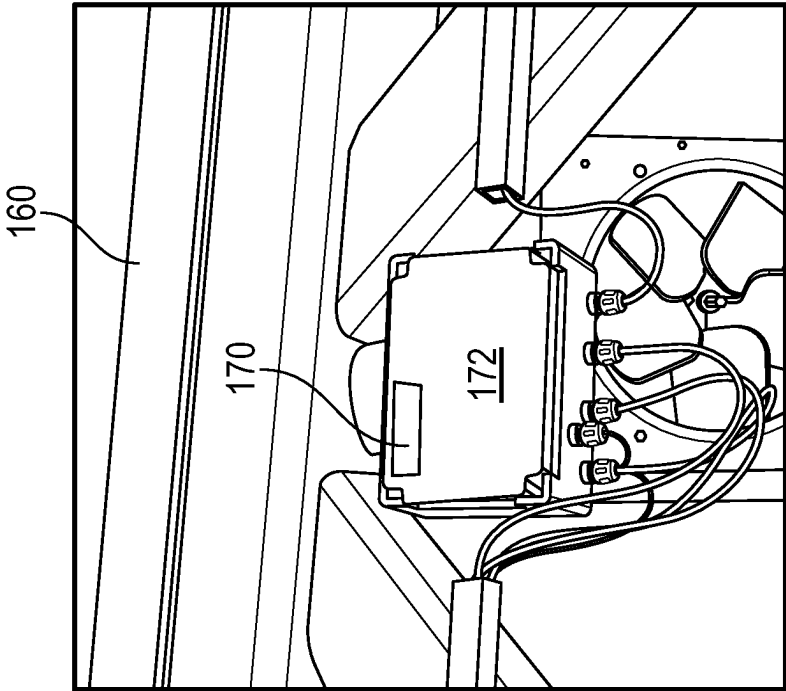


FIG. 21

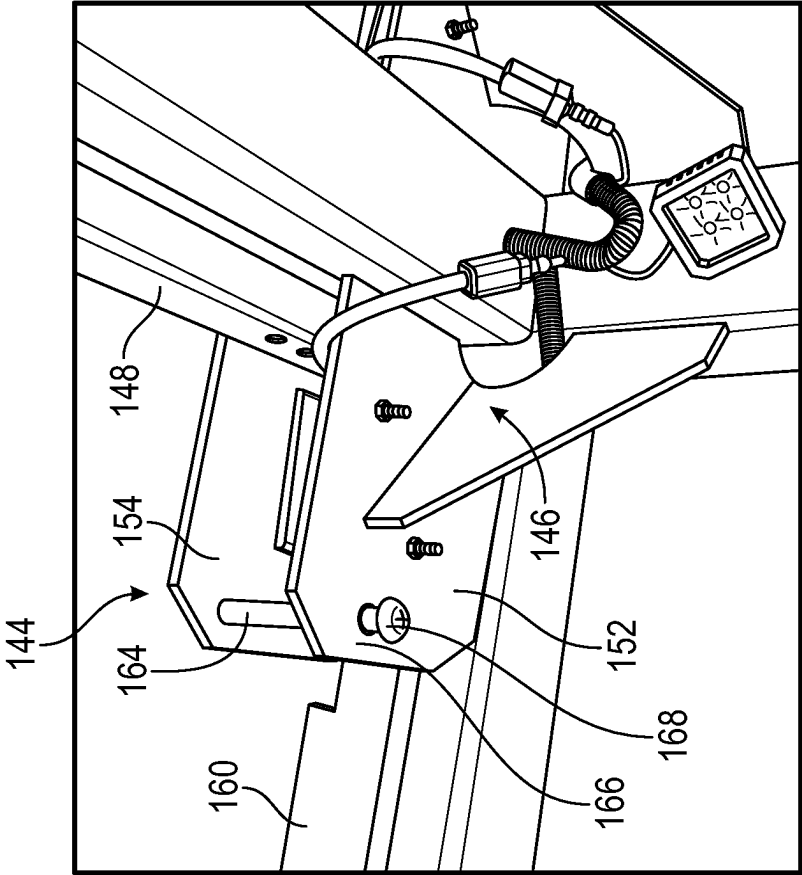


FIG. 20

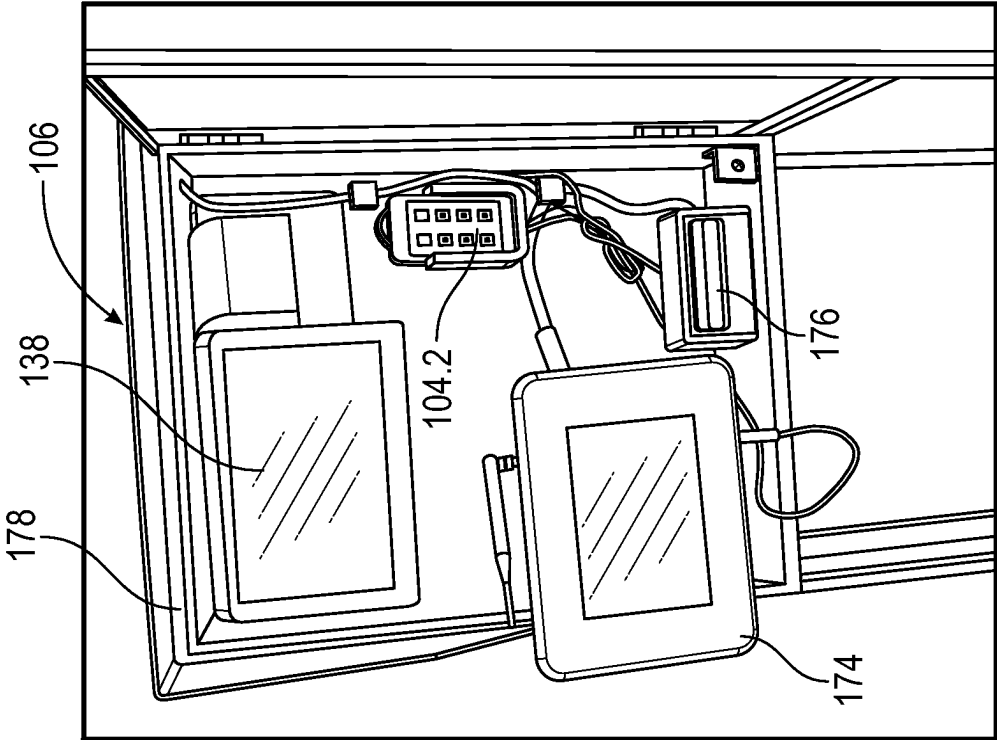


FIG. 23

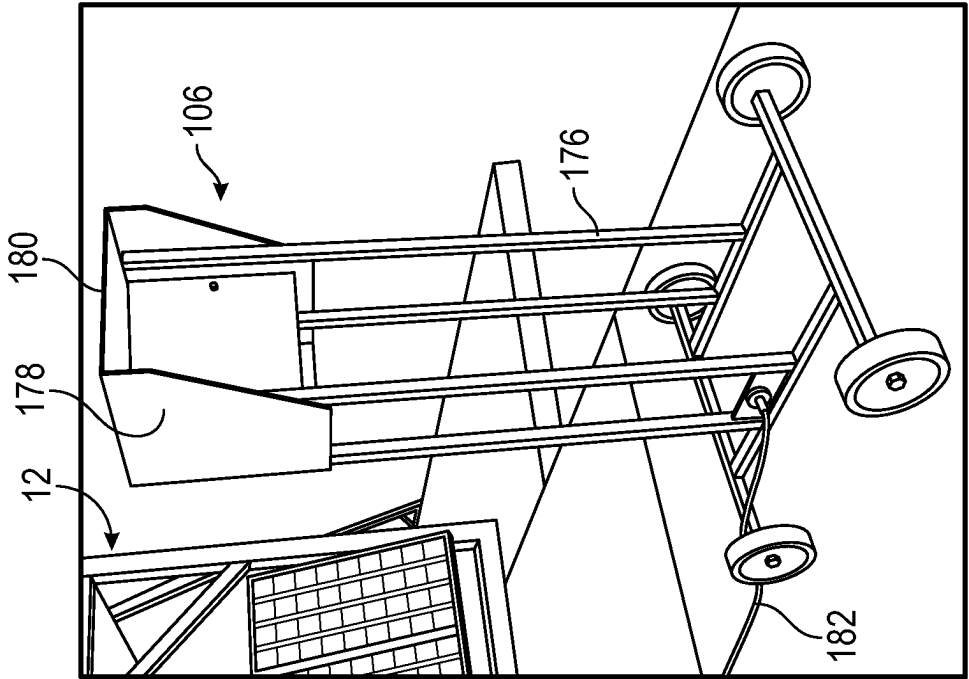


FIG. 22

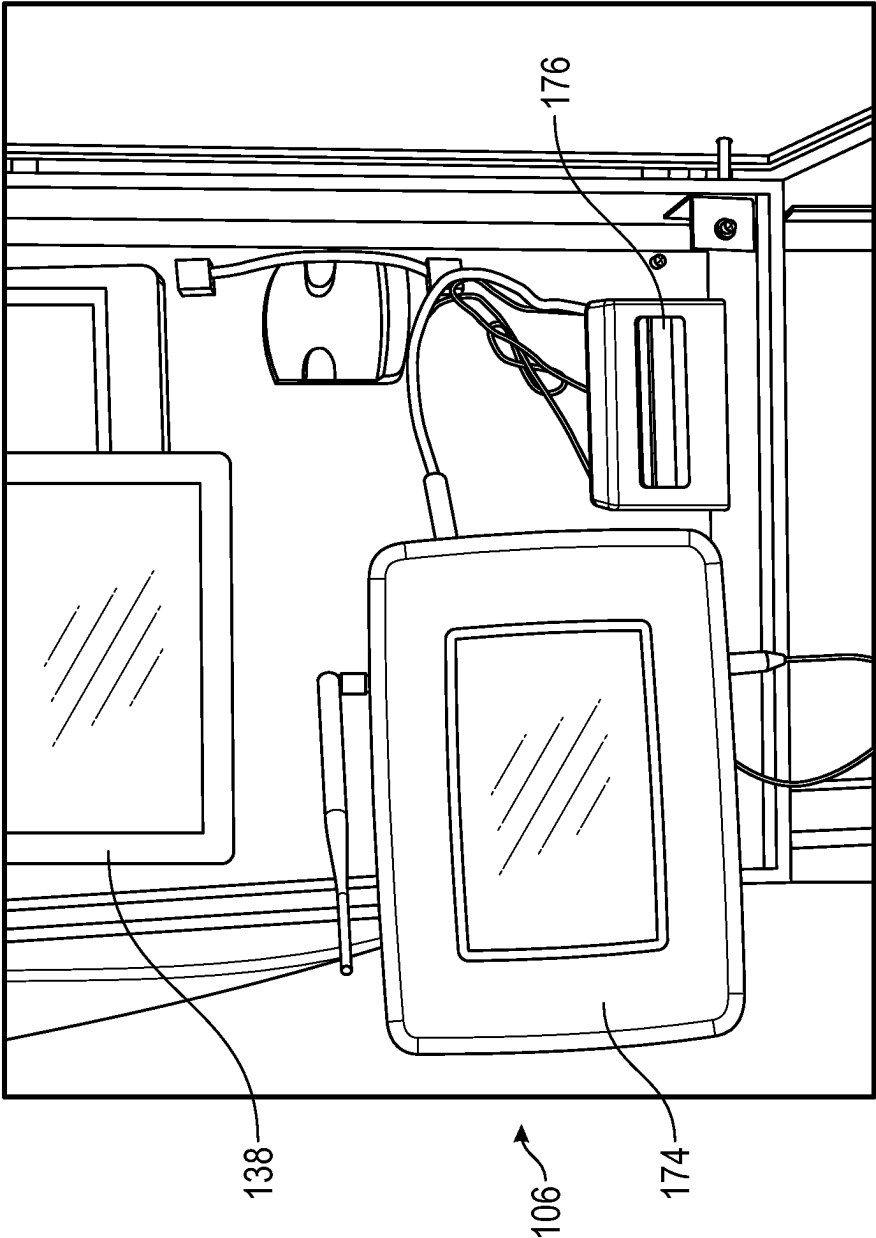


FIG. 24

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SUPPORT APPARATUS FOR PROPPANT STORAGE CONTAINERS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Division of application Ser. No. 17/086,050 filed on Oct. 30, 2020, which claims the benefit of U.S. Provisional Application No. 62/928,784, filed on Oct. 31, 2019. The entire disclosures of the above-referenced applications are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates a proppant discharge system for delivering proppant from a bulk storage container, and more particularly relates to a support apparatus configured to locate one or more modular proppant containers in an elevated position, deliver proppant to a feed station and accommodate various functional systems for efficient proppant delivery.

BACKGROUND

This section provides background information related to the present disclosure which is not necessarily prior art.

In the past several years, efforts have been made to improve logistics associated with the transportation, storage and delivery of proppant and other materials used onsite for fracturing operations at an oil/gas well, namely a fracturing site. In any hydraulic fracturing operation, a large amount of such proppant is required. Historically, it was been difficult to effectively transport and store the proppant at the fracturing sites. As a result, efforts have been made to load proppant into a modular container at a sand mine or transload facility, then transport the proppant-filled container to the fracturing site. Once onsite, the proppant-filled containers may be queued up at the fracturing site and proppant dispensed from the containers for use in the fracturing operation. Once emptied, the container may again be queued up at the fracturing site to transported back to the transload facility. Once proppant is material logistics and inventory management systems may be used to provide real-time, accurate information pertaining to the volume/inventory of proppant accessible to a user in a particular region or at a particular location.

Proppant conventionally used in fracturing operations must meet strict specification including moisture and turbidity requirements that require post-mining processes such as washing, screening and drying of the mined frac sand. Once so processed, proppant is relatively “slippery” and can be readily conveyed through handling equipment. Recent efforts to improve fracturing operations have focused on minimizing the post-mining processes of the frac sand by easing the specification for a suitable proppant. Therefore, there is a need to provide improved material handling equipment that is capable of conveying proppant having various characteristics.

SUMMARY

This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

A support apparatus for unloading a modular proppant container is disclosed herein. In one aspect the support apparatus includes a frame assembly having a base frame

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section, a plurality of posts extending upwardly from the base frame section and an upper frame section fastened to the plurality of posts in spaced relationship to the base frame assembly. The upper frame section provides an elevated load surface configured to support the modular proppant container in a position above a ground level. The base frame section includes a recessed region beneath the upper frame section providing a feed station. The support apparatus also includes a chute assembly supported by the frame assembly beneath the elevated load surface. The chute assembly includes a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet, and a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating above the recessed region. The support apparatus further includes a gate actuator having a coupling configured to engage with a gate assembly of the modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly.

In another aspect, the support apparatus a frame assembly includes an upper frame section and a plurality of posts fastened to the upper frame section in spaced relationship and extending downwardly therefrom. The upper frame section provides an elevated load surface configured to support the modular proppant container in a position above a ground level. The support apparatus also includes a chute assembly supported by the frame assembly beneath the elevated load surface. The chute assembly includes a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet, and a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating at a feed station below the elevated surface. The support apparatus further includes a gate actuator including a coupling configured to engage with the gate assembly of a modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly. The support apparatus further includes an on-board subsystem attached to the frame assembly for operating the support apparatus in a stand-alone mode. The on-board subsystem includes a controller configured to switch between a Sleep Mode in which the on-board subsystem is not operational and consumes little to no power and an On Mode in which the on-board subsystem is fully powered and operational.

In another aspect, the support apparatus includes a frame assembly having an upper frame section and a plurality of posts fastened to the upper frame section in spaced relationship and extending downwardly therefrom. The upper frame section includes two longitudinal beams and two cross beams forming a rectangular container bay configured to support the modular proppant container in a position above a ground level. The support apparatus also includes an in-situ weigh station configured to measure the weight of the modular proppant container supported on an elevated surface in the rectangular container bay. The in-situ weigh station includes a scale located at each corner of the rectangular container bay. Each scale has a base plate rigidly attached to the upper frame section and a load cell positioned on top of the base plate. The in-situ weigh station also include a weighing platform having an upper plate resting on top of the load cell at each corner of the rectangular

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container bay and a rectangular frame extending between the upper plate of adjacent load cells. The in-situ weigh station further includes a load cell processor configured to receive an input data signal from each load cell representing the vertical load between the base plate and the upper plate, compute a total weight on the in-situ weigh station and send an output data signal representing the total weight. The support apparatus further includes a chute assembly supported by the frame assembly beneath the elevated load surface. The chute assembly has a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet and a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating at a feed station below the elevated surface. The support apparatus additional includes a gate actuator having a coupling configured to engage with the gate assembly of a modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly.

Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

FIG. 1 is a front-left perspective view showing a support apparatus with a plurality of proppant containers located in an elevated position on the support apparatus;

FIG. 2 is a perspective view of a frame assembly for the support apparatus with FIGS. 2A-2C showing details of joint configuration at various locations (A-C) of the frame shown in FIG. 2;

FIG. 3 is a plan view of the frame shown in FIG. 2;

FIG. 4 is a front view of the frame shown in FIG. 2;

FIG. 5 is a right elevation of the frame shown in FIG. 2;

FIG. 6 is a detail view of a diagonal frame member;

FIG. 7 is a front perspective view of the support apparatus according to a first embodiment;

FIG. 8 is a rear perspective view of the support apparatus shown in FIG. 7;

FIG. 9 is a rear-right side perspective view of the support apparatus shown in FIG. 7;

FIG. 10 is a front-left perspective view of the support apparatus shown in FIG. 7

FIG. 11 is a rear-left perspective view of the support apparatus according to a second embodiment;

FIG. 12 is a front perspective view of the support apparatus shown in FIG. 11

FIG. 13 illustrates a main control panel with on-board systems of the support apparatus;

FIGS. 14-15 illustrate a wireless vision subsystem for monitoring the operation of the support apparatus;

FIGS. 16-17 illustrate an electrical grounding system for the support apparatus;

FIGS. 18-21 illustrate a weigh scale system for measuring the weight of a proppant container supported on the support apparatus; and

FIGS. 22-24 illustrate a remote operator console for monitoring and operating the support apparatus.

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Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

The following detailed description is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. Furthermore, there is no intention to be bound by any theory presented in the preceding background of the invention or the following detailed description.

With reference to FIG. 1, a plurality of proppant containers 10 are shown supported on a support apparatus 12 in an elevated position above ground level. The proppant containers 10 provide for the bulk storage of proppant or fracturing sand and are readily transportable to and from a well site. A presently preferred embodiment of each proppant container 10 is disclosed in U.S. Pat. No. 9,809,381 to Oren et al., which is expressly incorporated by reference herein. As Oren et al. describe therein, each proppant container includes a gate assembly which is selectively positionable between a closed position and an open position for metering proppant sand from the proppant container.

The support apparatus 12 includes a frame assembly 14 defining an elevated load surface 16 for the proppant containers 10 and a chute assembly 18.1, 18.2, 18.3 (collectively 18) located beneath for each of the proppant containers for directing proppant discharged from the proppant containers to a feed station. The feed station is a target site such as a blender hopper or proppant conveyor assembly for further handling of the proppant in the fracturing operation. One skilled in the art should recognize that the support apparatus 12 as described herein may be configured for one or more modular proppant containers. For example, in the embodiment illustrated in FIG. 1, the support apparatus 12 is configured with three bays or sections 12.1, 12.2, 12.3 for supporting three proppant containers 10.1, 10.2, 10.3 and therefore include three chute assemblies 18.1, 18.2, 18.3.

The support apparatus 12 also includes various on-board subsystems 20 associated with the frame assembly 14. The on-board subsystems 20 may include power supply subsystem having solar panels and/or battery banks as well as power conditioning circuitry for electrically powering the support apparatus, a hydraulic subsystem having a hydraulic pump, a fluid storage tank and a hydraulic for manipulating components of the support apparatus, and a vision subsystem for remotely monitoring the operational state of the support apparatus. These on-board subsystem also include subsystem controllers for operating the on-board subsystems 20 of the support apparatus 12. These on-board subsystems 20, which will be described in greater detail below, enable the support apparatus 12 to be used in a stand-alone or self-sufficient mode of operation that does not require separate or remote utilities such as an external power supply or external pneumatic or hydraulic power source. In this way, the support apparatus 12 physically supports modular containers for discharging proppant to a feed station and also operationally supports the functions of the process by providing electrical, pneumatic and hydraulic power for the support apparatus 12.

With reference now to FIGS. 2-6, the frame assembly 14 is fabricated using tubular steel members including vertical posts 30, longitudinal beams 32 and cross beams 34 welded together in a three-dimensional rectangular configuration. Diagonal beams 36 extend between the vertical posts 30 and the upper longitudinal beams 32.1, 32.2, 32.3 (collectively referred to as 32) as well as the vertical posts 30 and the

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upper cross beams **34.1**, **34.2**, **34.3** (collectively referred to as **34**) to enhance the rigidity of the frame assembly **14** by triangulating the joints between these tubular steel members.

An upper frame section includes upper longitudinal beams **32** and the upper cross beams **34** providing the elevated load surface **16** for supporting the proppant containers **10**. The lower longitudinal beams **32** and lower cross beams **34** define a base frame section or simply base **38** of the frame assembly **14**. The right and left bays **12.1**, **12.3** include a pair of intermediate longitudinal beams **401** extending between the lower cross beams **34** to provide additional floor support at the base of the support apparatus **12**. As illustrated in the figures, the middle bay **12.2** includes a pair of intermediate cross beam **40** secured to a top surface of the lower longitudinal beams **32.2** and extending therebetween. The intermediate cross beams **40c** are configured to receive a pair of forks extending from a fork lift vehicle for lifting and locating the support apparatus **12** at a well site. In another embodiment, the intermediate cross beams **40c** may be secured to a bottom surface of the upper longitudinal beams **32.2** in the middle bay **12.2**. A sheet material **42** may be fastened to the base **38** on the top of the beams **32**, **34**, **401**, **40c** forming a floor of the support apparatus **12**. As best seen in FIG. 3, the longitudinal beam **32.2** of the middle bay **12.2** is located inboard relative to the longitudinal beams **32.1**, **32.3** of the right and left bays **12.1**, **12.3** to provide a cut-out or recessed region **44** in the frame assembly **14**. This recessed region **44** is configured to receive a hopper for fracturing equipment such as a blender or similar equipment (not shown). In this regard, the recessed region provides a feed station for proppant being discharged from the proppant containers **10**. In other words, the frame assembly **14** has a recessed region **44** formed therein so that the support structure can be located at least partially above the feed station.

The frame assembly **14** may optionally include additional structural elements such as tabs **46** shown in detail **2A** and **2B** for attached items like sheet material **42** to the beams **32**, **34** or cross members **48** to provide additional rigidity and/or attachment locations for the on-board subsystems **20** of the support apparatus **12**. As illustrated in the figures, the sheet material **42** is a metal flooring grid, however one skilled in the art will appreciate that the sheet material **42** may be formed with other materials such as a plastic or composite material and/or be configured in a manner other than a flooring grid such as a diamond plate or other anti-slip sheet material. As presently preferred, the frame assembly **14** further includes a pair of outriggers **50** extending from one of the lower longitudinal beams **32** beyond the base **36**. For example, as best seen in FIGS. 8-9, the outriggers **50** extend from the rear lower longitudinal rail of right and left bays **12.1**, **12.3**. A leveling jack **52** is disposed at the end of each outrigger **50** and has a height adjustable foot **54** that can be positioned to stabilize and level the frame assembly in a stationary position. As illustrated, the leveling jacks **52** is hand operated to raising and lowering the foot **54** relative to the outrigger **50**. One skilled in the art should, however, recognize that the leveling jack may be motorized or hydraulically actuated for raising and lowering the foot **54**.

With reference to FIGS. 7-10, the chute assembly **18** and the on-board subsystems **20** will be further described. As previously explained, there is a chute assembly **18.1**, **18.2**, **18.3** associated with each bay **12.1**, **12.2**, **12.3** respectively. The various structure and function of each chute assembly is sufficiently similar that each chute assembly will be described by reference to chute assembly **18.1** shown in FIG. 8. The chute assembly **18.1** includes a funnel section

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56, a chute section **58** and a gate actuator **60**. The funnel section **56** is formed by a wall **62** tapering from an inlet or opening **64** at the top of the wall subjacent to the elevated load surface **16** to an outlet **66** below the inlet **64**. As illustrated, the funnel section **56** forms an inverted, truncated pyramid but may be formed by other similar shapes that provide a funneling function from the inlet to the outlet. The chute section **58** extends downwardly from a first end **68** at the outlet **66** of the funnel section **56** to a second end **70** opposite the first end **68** and terminating above the recessed region **44** of the frame assembly **14**. As illustrated, the chute section **58** includes an upper chute **72** secured to the funnel section and a lower chute **74** slidably supported on the upper chute **72**. An actuator **76**, in the form of a mechanical slide, is operably coupled between the upper and lower chutes **72**, **74** so that the overall length of the chute section **58** can be adjusted by sliding the lower chute **74** along the longitudinal axis of the upper chute **72**. One skilled in the art should recognize that the actuator **76** may take other forms such as a motor-driven slide, an electric ram, or a hydraulic or pneumatic or electric actuator. In these embodiments, the actuator **76** may include a limit switch for restricting the relative movement of the upper and lower chutes **72**, **74**.

FIGS. 11-12 illustrate an alternate embodiment of a support apparatus **12'** in which the chute section **58'** includes a cylindrical upper chute **72'** extending from the funnel section **56'** at the first end **68'** and a truncated conical lower chute **74'** extending from the upper chute **72'** and terminating at the second end **70'**. As illustrated, the length of the chute section **58'** is not adjustable, but an angular orientation of the chute section **58'** relative to the funnel section **56'** may be adjustable by providing a bearing or similar rotating interface at the first end **68'** between the funnel section **56'** and the chute section **58'**.

In some applications, proppant stored in the container **10** and dispensed with the support apparatus **12**, **12'** may have a higher moisture or turbidity than conventional proppant that has been dried in post-mining operations. It may be beneficial to specifically configure the support apparatus for these circumstances. For example, portions of the support apparatus **12**, **12'** may be fabricated using a stainless steel material or plastic for providing a slipperier surface than if fabricated using mild steel. In particular, the funnel section **56**, **56'** and/or the chute section **58**, **58'** may be fabricated from stainless steel or plastic. Alternately and/or additionally, the interior surfaces of these sections may be coated with a low friction coating such as a PTFE or similar non-stick coating for reducing the coefficient of friction of the interior surfaces.

Additional features may be used to promote gravity feeding proppant from containers **10** positioned on the support apparatus **12**, **12'** to the feed station. One such feature includes a shaker or vibration mechanism **184** operably coupled between the frame assembly **14**, **14'** and the funnel section **56**, **56'** and/or the chute section **58**, **58'** for gently vibrating these components as proppant discharged from the container **10**. For example, in an embodiment as shown in FIG. 11, the vibration mechanism **184** may include one or more actuators **186** operably coupled between the frame assembly **14'** and the funnel section **56'** for linearly, orbitally or rotationally vibrating the funnel section **56'** to shake loose any proppant that may have become lodged or stalled therein. While the vibration mechanism **184** is only illustrated in the center bay of frame assembly **14'**, one skilled in the art should appreciate that the other bays of frame assembly **14'** or frame assembly **14** may be similarly equipped with a vibration mechanism. Likewise, in an

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embodiment illustrated in FIG. 11, an actuator 188 may be operably coupled between the frame assembly 14' and the lower chute section 74' to shake loose any proppant that may have become lodged or stalled therein. While actuator 188 is only illustrated in the center bay of frame assembly 14, one skilled in the art should appreciate that the other bays of frame assembly 14 or frame assembly 14' may be similarly equipped with a vibration mechanism. Alternately, the actuator 76, which in reference to FIGS. 7 and 10 is used to adjust the length of the chute assembly 58, may also be actuated to vibrate the chute assembly 58 by moving the lower chute 74 relative to the upper chute 72. Actuators 186, 188 may be an electro-mechanical, pneumatic, or hydraulic component. In such applications, the funnel section 56, 56' and/or the chute section 58, 58' may be resiliently supported from the frame assembly 14, 14' to accommodate relative movement therebetween. The amount of shaking movement required may be determined based on the moisture content, turbidity and mesh size of the proppant being discharged from the support apparatus 12, 12'.

Another such feature may include an aeration mechanism 190 for injecting a quantity of compressed air or similar fluid stream into proppant being discharged from the container 10. For example, as shown in FIG. 12, one or more air injectors 192 coupled to a source of compressed air 194 may be located adjacent the funnel section 56' and/or in the chute section 58' for transporting proppant being discharged from the container in a fluid stream. In so doing, proppant exiting the container 10 is fluffed up to reduce the density of the transported material thereby reducing the likelihood of clumping or clogging of proppant. Alternately, a feed mechanism 196 may be implemented along the proppant transport path from the container 10 to the feed station. For example, the feed mechanism 196, represented by the broken line in FIG. 11 may be configured in the funnel section 56' of the support apparatus 12, 12' and/or along the chute section 58'. In an embodiment, the feed mechanism 196 is an auger device that moves proppant discharged from the container 10 into and through the funnel section 56, 56' and/or through the chute section 58, 58' to the feed station. In other embodiments, a conveyor belt, paddle wheel, air stream or similar devices may be implemented for transporting proppant from the funnel section 56' through the chute section 58'. While aeration mechanism 190 and the feed mechanism 196 are only illustrated in the center bay of frame assembly 14', one skilled in the art should appreciate that the other bays of frame assembly 14' or frame assembly 14 may be similarly equipped with an aeration mechanism or feed mechanism.

With continued reference to FIGS. 10-11 and 15, the gate actuator 60 is illustrated. The gate actuator 60 includes a coupling 78 supported on a slide or rail 80 positioned at or adjacent to the elevated load surface 16 extending over the funnel section 56, 56'. A linear actuator 82 is operably coupled between the frame assembly 14 and the coupling 78. As best seen in FIGS. 11 and 15, the coupling 78 is configured as a receptacle having a slot 84 formed in the top end thereof. The slot 84 receives a pin (not shown) on the gate assembly of a proppant container 10 supported on the elevated load surface. With the gate assembly coupled to the coupling 78, the actuator 82 functions as a drive mechanism for selectively positioning the coupling 78 (and its received pin) to adjust the gate assembly on the proppant container between the closed and opened position and meter proppant therefrom.

As mentioned above, the support apparatus 12 also includes various on-board subsystems 20 attached to the

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frame assembly 14. Referring to FIGS. 7-10 and 13, the support apparatus 12 is provided with power supply system 22 which include two battery banks 86 rigidly secured to the sheet material 42. The battery banks 86 include one or more batteries and power conditioning (not shown) that is electrically coupled to an electrical service panel 88 on a main control panel 90 supported at an end of the frame 14. The service panel 88 may be electrically coupled to the electrical components of the support apparatus 12 for providing primary or auxiliary electrical power thereto. With reference now to FIGS. 16-17, the power supply system 22 also includes an assembly 114 to electrically ground the support apparatus 12. The assembly 114 includes a grounding reel 116 with a retractable grounding wire 118 electrically connected to the support apparatus 12. The assembly 114 also includes a portable grounding rod 120 that can be readily pounded into the ground adjacent the support apparatus 12. A clamp or clip 122 attached to the end of the retractable ground with 118 can be releasably secured to the grounding rod 120 for electrically grounding the support apparatus 12.

The power supply system 22 may further include one or more solar panels 92 with power conditioning 94 electrically coupled to the battery banks 86 and/or the electrical service panel 88 to provide electrical power thereto for charging the battery banks 86 and/or for generating primary or auxiliary power. In one embodiment, the solar panels 92 are positionable with respect to the frame 14 between a stowed position and a deployed position. For example, as seen in FIG. 9, the solar panel 92 is pivotally supported along an upper edge 94 on an axle 96 between structural members of the frame 14 such as vertical posts 30 or diagonal beams 35. One or more supports 98 extend between the solar panel 92 and a lower portion of the frame 14. The supports 98 are adjustable for moving the solar panels 92 between a stowed position vertically oriented between the vertical posts 30 and a deployed position angularly extending from the frame 14 as shown in the figures. The supports 98 are adjustable to affect this pivoting movement of the solar panels 98. For example, the supports 98 may be a linear actuator for extending and retracting the length of the support. Alternately, the support 98 may be rotatably supported on the frame 14 and slidably supported on the solar panel 92. Latch posts 100 extend upwardly from the base 38 and terminate at a latch 102 that cooperates with the solar panel 92 for securing it in the stowed position. Once the latch 102 is released, the support 98, which in one embodiment is a pneumatic cylinder, may extend to deploy the solar panel 92.

One skilled in the art will recognize that the components used to deploy, operate and stow the various components of the support apparatus 12 may be manually adjustable (e.g., height adjustable foot 54), mechanically adjustable (e.g., chute actuator 76), electrically adjustable, hydraulically adjustable (e.g., gate actuator 82) or pneumatically adjustable (e.g., solar panel supports 98). When implementing an electrically adjustable component, such as an electrical actuator, it is electrically coupled to the electrical service panel 88 via control devices 104.1, 104.2, which may be located locally on the main control panel 90 or remotely in a remote operator console 106. When implementing a hydraulically adjustable component, such as a hydraulic actuator, it is hydraulically coupled to the other components of the hydraulic system including a hydraulic pump 100 in fluid communication with a hydraulic storage tank or sump 112 and a hydraulic system controller 108.

The on-board subsystems 20 may include a vision subsystem 124 configured to visually monitor the state and operational status of the support apparatus 12. As shown in

FIG. 14, the vision subsystem 124 includes a camera assembly 126 for monitoring proppant being discharged from the chute assembly 18. Camera assembly 126 includes a U-shaped support brace 128 extending from the upper chute 72.2 of the middle chute assembly 18.2. A camera 130 is secured to the support brace 128 and aimed toward the recessed region 44 in the frame so that the second ends 70 of the lower chutes 74 and a hopper (not shown) positioned in the feed station are within a field of view of the camera 130. As shown in FIG. 15, the vision subsystem 124 also includes a camera assembly 132 for monitoring the position of the gate actuator 60 in each of the container bays. Camera assembly 132 includes a support beam 134 extending between upper cross beams 34. A camera 136 is secured to the support beam 134 and aimed toward the gate actuator 60 so that the coupling 76 and the end of the linear actuator 82 are within a field of view of camera 136. In an embodiment, camera 130, 136 are weatherproof cameras electrically coupled to the power supply system 22 and configured to capture and wirelessly transmit live video from a low light scene to a remote display 138 (see FIGS. 23-24). One skilled in the art will recognize that additional camera assemblies may be deployed on and around the support apparatus 12 to visually monitor the state and operational status of the support apparatus 12.

The on-board subsystems 20 may include an in-situ weigh scale system 140 configured to measure the weight of the containers 10 supported on the support apparatus 12. As shown in FIGS. 8 and 18-21, the weigh scale system 140 includes an in-situ weigh station 142.1, 142.2, 142.3 for each of the container bays 12.1, 12.2, 12.3. It will be understood that each of the in-situ weigh stations 142 are substantially the same such that only one weigh station needs to be further described herein. The in-situ weigh station 142 includes four scales 144, one located at each of the corner of the container bay 12. With particular reference now to FIGS. 18-21, each scale 144 includes a lower support or base 146, a weighing platform 148 and a load cell 150 positioned between the base 146 and the weighing platform 148. The base 146 includes a base plate 152 rigidly secured between the upper longitudinal beams 32 and upper cross beams 34, and a gusset plate 154 rigidly secured between the base plate 152 and the vertical post 30. The load cell 150 is secured to the base plate 152 by bolts or another similar fastening technique.

The weighing platform 148 includes an upper plate 154 resting on top of the load cell 150. A stacking cone 156 may be welded to the top of the upper plate 154 and configured to engage, locate and stabilize a container 10 loaded into the bay. A rectangular frame 158 includes angle iron members 160 having a horizontal flange that extends above the upper longitudinal beams 32 and the cross beams 34 and a vertical flange set inside the upper longitudinal beams 32 and the cross beams 34 between the upper plate 154 of adjacent load cells 150 within a given bay. In this way, the weighing platform 148 floats on top of the four scales 144 within the in-situ weigh station 142. A retainer extends from the weighing platform 148 and is configured to prevent the upper plate 154 from lifting off of the load cell 150 when a container is removed from the support apparatus 12. In an embodiment shown in FIGS. 18-19, the retainer is a J-shaped catch 162 extending downward from the frame 158 beneath the upper longitudinal and cross beams 32, 34 to impede any significant upward displacement of the weighing platform 148. In an embodiment shown in FIG. 20, the retainer is a pin 164 extend downwardly from the bottom of the upper plate 154 and through a hole 166 in the base plate 152. A stop 168 is formed on an end of the pin 164 opposite

the upper plate 154 to impede any significant upward displacement of the weighing platform 148.

Each of the load cells 150 in a given weigh station 142 generates a data signal based on the vertical load (i.e., weight) between the base 146 and the weighing platform 148. These data signals are communicated to a load cell processor 170 in a junction box 172 that computes a total weight on a given weigh station 142, which may be displayed locally at the weigh station 142 or communicated to a remote digital readout 174 (see FIGS. 23-24).

With reference now to FIGS. 22-24, the support apparatus 12 may include a remote operator console 106 which is in wired and/or wireless communications with the other subsystems 20 previously described. In the embodiment illustrated in FIGS. 22-24, the remote operator console 106 includes a wheeled stand or dolly 176 supporting an enclosure 178 having a hood 180. An umbilical cord 182 extends from the console 106 to the support apparatus 12 and may include electrical cords for providing electrical power to the enclosure 178 and/or data transmission cords for communicating data signals and control signal between the support apparatus 12 and the remote console 106. As shown in FIGS. 23-24, the remote console includes a video display 138 for the vision subsystem 124, the remote digital readout 174 for the weigh scale system 140, a tablet 104.2 configured to remotely control the hydraulic system controller 108 and an on-board power supply 176. One skilled in the art will recognize that the remote operator console may be equipped with additional systems used in the support of proppant delivery to a target site at a fracturing operation such as a radio for providing two-way communication, a laptop computer or other local computing device.

When deployed for a fracturing operation near an oil well site, the support apparatus 12 provides an efficient means for establishing the necessary infrastructure to deliver proppant to a hopper in the feed station. In this regard, the support apparatus 12 is located in a predetermined location using a field transport vehicle such as a fork lift truck. Specifically, the support apparatus 12 may be transported to the well site using conventional means such as a rail car or flatbed trailer-truck. The forks of the fork lift truck are positioned into cross beams 38 and the support apparatus 12 is removed and placed at the predetermined location. For example, the support apparatus 12 may be placed next to a blender apparatus such that the blender hopper (not shown) is located in the recessed region 44 of the frame 14. Next the support apparatus 12 may be stabilized by extending the feet 54 from the leveling jacks 52 at the end of the outriggers 50 to level the support apparatus 12. The chutes 18 are then positioned so that the second end 72 of the chute section 58 is above the blender hopper. The solar panels 92 may also be moved from the stowed position to the deployed position and the on-board subsystems 20 powered up and tested for proper functioning.

Once so positioned, the field transport vehicle may be used to retrieve a proppant container 10 and locate it on the elevated load surface 18 such that the pin of a gate assembly is received in the coupling 78. Additional proppant containers 10 may be retrieved and located on the elevated load surface 18 until all the bays of the support apparatus 12 are occupied. The gate actuator 60 is remotely operated to selectively position the gate assembly for each proppant container 10. Proppant for each container 10 is gravity-fed through the chute assembly 58 for delivery to the blender hopper. When a proppant container 10 has emptied all of the proppant stored therein, the empty proppant container 10

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may be removed from the support apparatus 12 and replaced with a filled proppant container 10.

Because the support apparatus 12 can be operated in a stand-alone mode, the various controllers associated with the on-board subsystems 20 can be configured to reduce power consumption by switching between a "SLEEP" mode in which subsystems are not operational and consume little to no power and an "ON" mode in which the subsystems are fully powered and operational. For example, the hydraulic system and particularly the hydraulic pump 100 consumes a significant amount of power when fully powered and operational. As such, the hydraulic system controller 108 may be configured to go into SLEEP mode by shutting down the hydraulic pump 100 when hydraulic pressure is not needed to operate a hydraulic component. Once a control signal is received to operate a hydraulic component, for example when manipulation of the gate actuator 60 is requested, the hydraulic system controller 108 switches to the ON mode and turns on the hydraulic pump 100 to provide hydraulic pressure for operating the hydraulic component. In this way, the hydraulic system may be instantaneously activated to provide on-demand hydraulics, while conserving power when hydraulics are not needed. A similar on-demand activation may be implemented for other subsystems associated with the support apparatus 10.

Various embodiments and methods have been presented in the foregoing detailed description, it should, however, be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the invention in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment as contemplated herein. It should be understood that various changes may be made in the function and arrangement of elements described in an exemplary embodiment without departing from the scope of the invention as set forth in the appended claims.

What is claimed is:

1. A support apparatus for unloading a modular proppant container having a gate assembly comprising:

a frame assembly including an upper frame section having a pair of and a plurality of posts fastened to the upper frame section in spaced relationship and extending downwardly therefrom, the upper frame section having two longitudinal beams and two cross beams forming a rectangular container bay configured to support the modular proppant container in a position above a ground level;

an in-situ weigh station configured to measure the weight of the modular proppant container supported on an elevated surface in the rectangular container bay, the in-situ weigh station including:

a scale located at each corner of the rectangular container bay, each scale including a base plate rigidly

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attached to the upper frame section and a load cell positioned on top of the base plate;

a weighing platform including an upper plate resting on top of the load cell at each corner of the rectangular container bay and a rectangular frame extending between the upper plate of adjacent load cells; and
a load cell processor configured to receive an input data signal from each load cell representing the vertical load between the base plate and the upper plate, compute a total weight on the in-situ weigh station and send an output data signal representing the total weight;

a chute assembly supported by the frame assembly beneath the elevated load surface, the chute assembly having a funnel section formed by a wall tapering from an inlet at a top of the wall subjacent to the elevated load surface to an outlet below the inlet, a chute section extending downwardly from a first end at the outlet of the funnel section to a second end opposite the first end and terminating at a feed station below the elevated surface; and

a gate actuator including a coupling configured to engage with the gate assembly of a modular proppant container supported on the elevated load surface and a drive mechanism extending between the frame assembly and the coupling to selectively position the coupling for adjusting the gate assembly.

2. The support apparatus according to claim 1, wherein the rectangular frame of the weighing platform comprises a horizontal flange extending above the longitudinal beams and the cross beams and a vertical flange set inside the upper longitudinal beams and the cross beams, wherein the weighing platform floats on top of the four scales within the in-situ weigh station.

3. The support apparatus according to claim 1, wherein the in-situ weigh station further comprises a plurality of retainers extending from the weighing platform and configured to engage the upper frame section for preventing the upper plate from lifting off of the load cell when the modular proppant container is removed from the support apparatus.

4. The support apparatus according to claim 3, wherein each retainer comprises a J-shaped catch extending downward from the rectangular frame beneath at least one of the longitudinal beam and the cross beams to impede upward displacement of the weighing platform when the modular proppant container is removed from the support apparatus.

5. The support apparatus according to claim 3, wherein each retainer comprises a pin extending downwardly from a bottom of the upper plate and through a hole formed in the base plate and a stop formed on an end of the pin opposite the upper plate to impede any significant upward displacement of the weighing platform when the modular proppant container is removed from the support apparatus.

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